VENDOR ONBOARDING AND COMPLIANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

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BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND BUSINESS SYSTEMS



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I, Nithiya Sree N C (241401060), hereby declare that the project report titled "Vendor Onboarding and Compliance Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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ABSTRACT

Organisations must collect vendor details, verify supporting documents, carry out compliance checks and approve or reject vendors before they are engaged. When these activities are handled through spreadsheets, e-mails and separate folders, vendor information becomes inconsistent, reviews are delayed, nothing links a decision to the documents and checks behind it, and the expiry dates of compliance documents are easily missed.

The Vendor Onboarding and Compliance Management System is a low-code application built on Zoho Creator to centralise this process. Low-code suits the problem because vendor onboarding is a data-entry, review and notification workflow that the platform's forms, reports and dashboard components cover directly. The application (named Vendor Compliance System in the Creator console) contains four linked forms – Vendor Registration, Compliance Documents, Compliance Checks and Approval Records – joined by Vendor lookup fields. Deluge workflows generate a unique Vendor ID, set the initial vendor status to Submitted, set uploaded documents to Pending, update the vendor status after an approval decision, send e-mail notifications to the Compliance Officer and to vendors, and run a daily scheduled workflow that reminds vendors of documents expiring within 30 days. The vendor status moves through Submitted, Under Review, Approved, Rejected and Needs Correction. Reports include list reports for all vendors, pending applications and document verification, an expiring-documents calendar, an Approval Pipeline Kanban board and a compliance overview chart, and a dashboard combines KPI tiles, charts, recent applications and quick actions. Access is organised for four roles – Administrator, Vendor, Compliance Officer and Procurement Manager – so that vendors reach only their own records.

The application was tested through its live interface with sixteen functional test cases, all of which passed, and six defects found during development were corrected and are recorded with their root causes. Zoho Mail notification workflows are the planned integration; no REST API call is used. The report also records the limitations that remain – the live role configuration, record-level vendor access, deployment across environments and security testing are only partly evidenced – and lists them as future enhancements.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Vendor Onboarding; Compliance Management; Workflow Automation

PROJECT AT A GLANCE

Attribute	Details
Project Title	Vendor Onboarding and Compliance Management System
Application Domain	Vendor management and compliance management
Student Name & Register No.	NITHIYA SREE N C – 241401060
Zoho Creator Edition	Trial edition (application header shows “Trial expires in 14 days”)
Application Name (Link Name)	Vendor Compliance System (vendor-compliance-system)
Published / Live URL	
Demo Credentials (per role)	Not printed in this report
Number of Forms / Subforms	4 (Vendor Registration, Compliance Documents, Compliance Checks, Approval Records) / 0
Number of Reports (by type)	List: All Vendors, Pending Applications, Document Verification · Calendar: Expiring Documents · Kanban: Approval Pipeline · Chart: Compliance Overview (6 in all)
Number of Pages / Dashboards	1 (Vendor Compliance Dashboard with KPI tiles, charts, recent applications and quick actions)
Number of Workflows	Form: 6 (Vendor ID generation, DS-01 to DS-04 and Alert on High Risk Vendor) · Schedule: 1 (DS-05, daily) · Approval, Blueprint, Batch: not configured as separate components
Deluge Functions / Custom Actions	Deluge scripts inside form workflows and one scheduled workflow; no stand-alone function
Integrations	Zoho Mail notification workflows (planned integration); no REST API
Roles & Profiles Defined	Roles: Administrator, Vendor, Compliance Officer, Procurement Manager ·
Environments Used	Development (trial edition)
Date of Production Deployment	02.11.2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
DS	Deluge Script (script ID used in this report)
ER	Entity–Relationship
FR	Functional Requirement
GSTIN	Goods and Services Tax Identification Number
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
PAN	Permanent Account Number
PIN	Postal Index Number
RBAC	Role-Based Access Control
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

Every organisation that buys goods or services depends on its vendors. Before a vendor is engaged, the organisation normally collects the vendor's company and contact details, checks supporting documents such as incorporation and tax certificates, carries out compliance checks, and decides whether the vendor may be approved. In many organisations these steps are carried out through spreadsheets, e-mail threads and separate folders. A vendor's details are then held in one place, its documents in another and the reviewer's opinion in a third, so nobody can see the whole position of a vendor at a glance.

The Vendor Onboarding and Compliance Management System is a low-code application built on Zoho Creator to replace this manual practice with one structured system. A vendor registers by submitting company and contact information, including PAN and GSTIN. Compliance documents are uploaded against the vendor with an issue date and an expiry date. The Compliance Officer verifies each document, records compliance checks with a risk level and remarks, and the Procurement Manager records the final decision – approved, rejected or needs correction – which updates the vendor status. Workflows send e-mail notifications to the Compliance Officer and to the vendor, and a daily scheduled workflow reminds vendors of documents that will expire within 30 days.

The application is named Vendor Compliance System in the Creator console. It contains four linked forms, six reports, one dashboard, Deluge workflows and role-based access for four roles. The vendor status is maintained throughout the onboarding process as Submitted, Under Review, Approved, Rejected or Needs Correction (Section 4.4).

1.2 Problem Statement

Manual vendor onboarding gives neither the organisation nor the vendor reliable information about the state of an application. A low-cost digital solution is required that captures vendor details in a structured form, stores each vendor's compliance documents with their dates, records every document verification and compliance check, captures the approval decision with its reviewer and comments, keeps the vendor status up to date automatically, informs the reviewer and the vendor by e-mail and presents the position on a single dashboard. The main pain points are:

- Vendor details, documents and review notes are kept in different files, so information becomes inconsistent.

- Reviews are delayed because reviewers are not told when an application or a document arrives.
- Documents whose expiry dates are missed leave the organisation dealing with vendors whose compliance has lapsed.
- Decisions are not traceable to the checks and documents behind them, and vendors cannot see the status of their own application.
- Without access control, internal review information may be visible to vendors, and one vendor may see another vendor's records.

1.3 Objectives

The project set out to achieve the following objectives:

- To design a data model of four linked forms – Vendor Registration, Compliance Documents, Compliance Checks and Approval Records – with a Vendor lookup from each of the last three forms to the vendor record.
- To automate the generation of a unique Vendor ID, the initial vendor status and the initial verification status of uploaded documents using Deluge workflows.
- To record document verification, compliance checks, risk levels, reviewer details and remarks so that every review is traceable.
- To record approval decisions – approved, rejected or needs correction – and to update the vendor status from them.
- To notify the Compliance Officer and vendors by e-mail and to send daily reminders for documents expiring within 30 days, using Zoho Mail notification workflows.
- To provide reports and a dashboard that show vendors, pending applications, document verification, expiring documents, the approval pipeline and compliance activity.
- To control access through four roles so that vendors see only their own records and cannot reach internal reports or approval functions.

1.4 Scope of the Project

In scope. The application contains four forms (Vendor Registration, Compliance Documents, Compliance Checks and Approval Records), six reports (All Vendors, Pending Applications, Document Verification, an Expiring Documents calendar, an Approval Pipeline Kanban board and a Compliance Overview chart), one dashboard, Deluge workflows for status updates and e-mail notifications, a daily scheduled workflow for expiring documents, a vendor status process and a role design for administrators, vendors,

compliance officers and procurement managers. E-mail notifications use Zoho Mail notification workflows as the planned integration.

Out of scope. The following are not part of the application and are not claimed in this report: a separate REST API or Postman integration, Zoho Analytics, AI-based risk prediction, Zoho Catalyst, custom widgets and integrations with other services.

1.5 Justification for a Low-Code Approach

A vendor onboarding application is a data-entry, review and notification application: a few forms, a handful of reports, simple status changes and e-mails. These are exactly the building blocks that a low-code application platform provides as configurable components. On Zoho Creator the forms, their lookup relationships, the reports, the Kanban board and the dashboard were assembled with the form builder, report builder and page builder, and only the status updates, the notifications and the expiry reminder required scripting in Deluge. Hosting, authentication, user management, responsive rendering and record storage are supplied by the platform, so the project effort could be concentrated on the onboarding and compliance logic rather than on infrastructure. Published studies of low-code practice report that drag-and-drop development and out-of-the-box components speed up development and that low-code is most favoured for process- and workflow-centred applications, although practitioners also report limits in customisation [4], [5].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing arrangement for onboarding a vendor is manual. The vendor sends its company details and documents by e-mail or on paper. A purchasing or compliance officer copies the details into a spreadsheet, saves the documents in a shared folder and notes the progress of the review in e-mails or in the spreadsheet. After the checks, a manager informs the vendor of the outcome by e-mail. Expiry dates of documents are tracked, if at all, by a reminder in a personal calendar. Figure 2.1 summarises this commonly followed practice. The description is a general account of manual vendor onboarding and is not a survey of one named organisation.

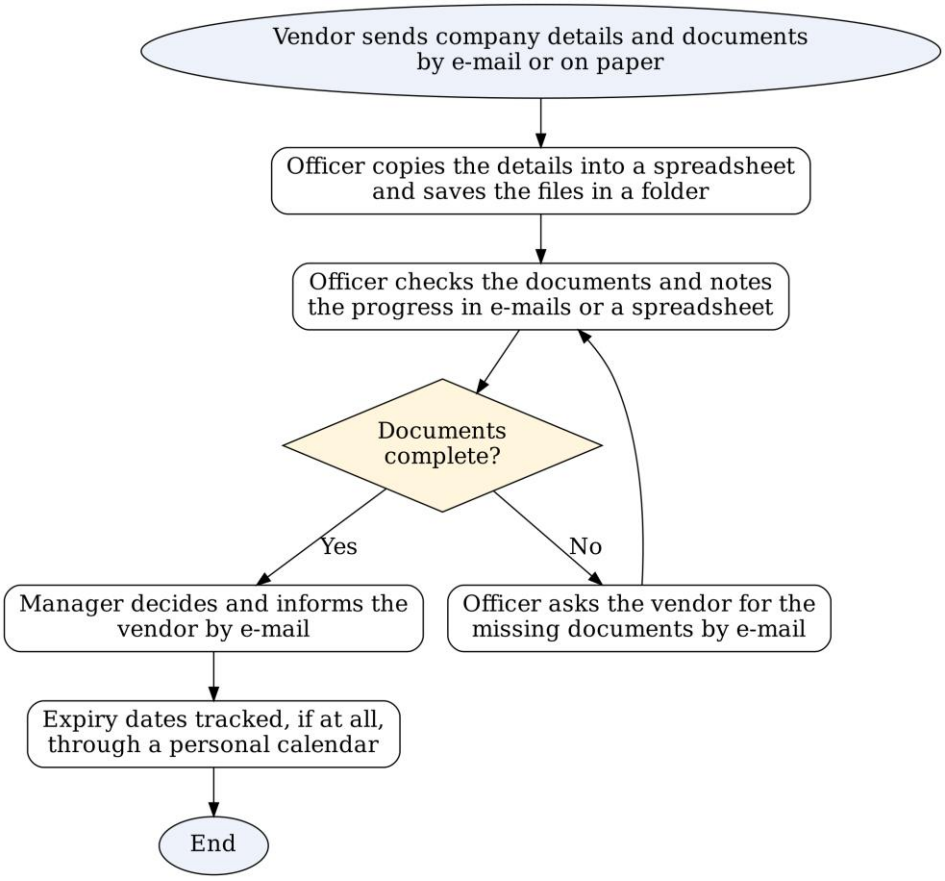


Figure 2.1: As-Is Process Flow of Manual Vendor Onboarding

2.2 Limitations of the Existing System

- **Inconsistent records.** Vendor details are copied between e-mails, spreadsheets and folders, so the same vendor may appear with different details in different places.
- **Delayed reviews.** Reviewers learn of a new application or document only when they

open the mailbox or the folder.

- **Missed expiry dates.** Nothing alerts the organisation or the vendor when a certificate is about to expire.
- **No audit trail.** The documents, checks and decision behind a vendor's status are not linked to each other, so a decision cannot easily be traced.
- **No consolidated view.** Pending applications, approved vendors and documents due to expire must be counted by hand.
- **No access control.** Files and spreadsheets cannot separate what a vendor, a reviewer and a manager may see.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a traditional pro-code implementation of the vendor onboarding system with the low-code implementation on Zoho Creator. The comparison is qualitative; no development-time or cost measurements were recorded during the project, so none are claimed.

Table 2.1: Comparison of Low-Code and Traditional Development for the Vendor Onboarding and Compliance Management System

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Database design, back-end services, user interface and deployment pipeline must each be built and integrated.	Forms, reports and the dashboard are assembled in builders; Deluge was needed only for status, notification and reminder logic.
Skills required	Programming in a front-end and a back-end language, SQL, hosting and security administration.	Platform familiarity and basic Deluge scripting.
Hosting and infrastructure	Servers or cloud services must be provisioned, secured and maintained.	Hosted and maintained by the platform.
File storage	A file store, upload handling and virus and size checks must be built.	A File Upload field stores documents with the record.
Notifications and scheduling	E-mail servers, job schedulers and error handling must be coded.	E-mail actions are configured in workflows, and a scheduled workflow runs daily.
Security and access control	Authentication, sessions, role logic and record sharing must be implemented in code.	Users, roles, permission profiles and record-visibility rules are configured in settings.
Maintenance	Changes require code changes, testing and redeployment.	Changes are made in the builder and published from Development to Production.

2.4 Literature Survey

Table 2.2 summarises the sources reviewed. Two are empirical studies of low-code practice;

three are the vendor documentation sets for the platform services on which the application depends. Only findings that could be confirmed from the cited pages are recorded.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, 2021 [4]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analysis of Stack Overflow and Reddit posts found that drag-and-drop interfaces and out-of-the-box components make low-code platforms easy to learn and speed up development, that they are favoured for automated processes and workflows, and that practitioners hold conflicting views of their advantages and disadvantages.	Supports the choice of a low-code platform for a workflow-oriented onboarding application and motivates the limitations discussion in Section 8.4.
2	M. A. Al Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider and A. Iqbal, 2021 [5]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges	Stack Overflow posts on nine low-code platforms identified 13 discussion topics in four categories; customisation was the largest category, and dynamic event handling was both the most popular and the hardest topic to obtain accepted answers for.	Highlights event handling and customisation as risk areas, which corresponds to the form-event workflows and Deluge scripts used in this application.
3	Zoho Corporation [1]	Zoho Creator Resource Center	Central documentation for building applications, using Deluge, managing users and assigning roles and permissions.	Reference for forms, reports, pages, workflows and user permissions.
4	Zoho Corporation [2]	Introduction to Deluge (Zoho Deluge Help)	Describes Deluge as Zoho's scripting language used to manipulate data, for example by creating or updating records.	Reference for the status-update, notification and expiry-reminder scripts.
5	Zoho Corporation [3]	Zoho Mail Help Center	Documentation for Zoho Mail, the e-mail service, including its use with other Zoho applications.	Reference for the planned Zoho Mail notification workflows.

2.5 Emerging Trends in Low-Code Development

Three general trends are relevant to this project. First, low-code platforms increasingly embed AI assistance in their builders; the Zia assistant icon is visible in the Zoho Creator builder screens captured for this project, although no AI feature was used in the application. Second, platforms emphasise integration with the surrounding service ecosystem, which this application exercises through Zoho Mail notification workflows. Third, as more business users build applications, governance concerns such as role design, record-level access, data privacy settings and environment management become more important; this application addresses them with four roles, permission profiles and vendor-specific record access. The

empirical literature also records that customisation limits and event handling remain recurring difficulties for developers [4], [5]. These trends are stated qualitatively and are not backed by market figures.

2.6 Proposed System

The proposed system is the Vendor Onboarding and Compliance Management System. A vendor registers by submitting its company, contact, address, tax and vendor-type details; the application generates a unique Vendor ID and sets the status to Submitted, and the Compliance Officer is notified. Compliance documents are uploaded against the vendor with their issue and expiry dates and are set to Pending; the Compliance Officer is notified of each upload, verifies or rejects each document with remarks, and the vendor is notified of the result. The Compliance Officer records compliance checks with a result and a risk level. The Procurement Manager records an approval decision – Approved, Rejected or Needs Correction – and the vendor status is updated from it. A daily scheduled workflow reminds vendors of documents that expire within 30 days. Reports and a dashboard present the vendors, the pending applications, the documents and the compliance activity, and role-based access ensures that vendors see only their own records. The proposed system therefore replaces the spreadsheets and e-mail threads with one traceable record for each vendor.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The roles below correspond to the four roles defined for the application.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Owner of the application configuration.	Manage application configuration, users, permissions and records.
Vendor	External user whose company is being onboarded.	Register, maintain permitted profile details, upload documents and view own status.
Compliance Officer	Internal reviewer of applications and documents.	Review applications, verify documents, conduct compliance checks and request corrections.
Procurement Manager	Internal decision maker.	Review compliance results and approve, reject or request corrections.
E-mail service (system actor)	Zoho Mail notification workflows used by the application.	Delivers the notifications to the Compliance Officer and to vendors.

3.2 Functional Requirements

Table 3.2 lists the functional requirements. FR-01 to FR-09 correspond to features of the application. Vendors must be able to access only their own records, and internal reports and approval functions must not be accessible to vendors.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall capture vendor registrations with Vendor ID, Company Name, Contact Person, Email, Mobile Number, Address, State, PIN Code, Vendor Type, PAN, GSTIN, Registration Date and Vendor Status; shall generate a unique Vendor ID; shall set the status to Submitted after registration; and shall validate mandatory fields, mobile number and PAN format.	Vendor, Administrator	High
FR-02	The system shall store compliance documents against a vendor with Document Type, File Upload, Issue Date, Expiry Date, Verification Status and Reviewer Remarks; shall set uploaded documents to Pending; and shall allow a document to be marked Verified or Rejected.	Vendor, Compliance Officer	High
FR-03	The system shall record compliance checks with Vendor, Check Type, Result, Risk Level, Reviewer, Check Date and Remarks.	Compliance Officer	High
FR-04	The system shall record approval decisions (Approved, Rejected, Needs Correction) with reviewer, comments and decision date, and shall update the vendor status from the decision.	Procurement Manager	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall provide the reports All Vendors, Pending Applications, Document Verification, an Expiring Documents calendar, an Approval Pipeline Kanban board and a Compliance Overview chart.	Administrator, Compliance Officer, Procurement Manager	High
FR-06	The system shall provide a dashboard with the KPIs Total Vendors, Pending Approvals, Approved Vendors and Documents Expiring Within 30 Days, a compliance chart, Recent Applications and Quick Actions.	Administrator, Compliance Officer, Procurement Manager	Medium
FR-07	The system shall send e-mail notifications: new registration and new document upload to the Compliance Officer; document verification result, approval or rejection decision and document expiry reminder to the vendor.	System	High
FR-08	The system shall restrict access by role so that vendors see only their own records and cannot reach internal reports or approval functions.	Administrator	High
FR-09	The system shall run a daily scheduled workflow that reminds vendors of documents expiring within 30 days.	System	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall open without perceptible delay for the record volumes of the demonstration. No formal timing measurement was recorded.
NFR-02	Security	Users shall reach only the forms, reports and pages permitted by their role; vendors shall not see other vendors' records or internal reports.
NFR-03	Usability	A vendor shall be registered with one form submission, and the application shall run in a standard web browser.
NFR-04	Availability	The application relies on the availability of the Zoho Creator service; no separate hosting is maintained by the project.
NFR-05	Maintainability	Business rules shall be held in named workflows so that they can be reviewed and changed without redesigning the forms.
NFR-06	Traceability	Each document, compliance check and approval record shall refer to its vendor and carry its reviewer, remarks and dates.
NFR-07	Reliability of notifications	A notification shall be produced for each registration, document upload, verification result and decision, and a daily reminder for each document expiring within 30 days.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition indicated in the application header), Development environment
Scripting language	Deluge (Zoho's scripting language for Creator workflows and functions)

Category	Specification
Browser	A current desktop web browser; the builder also offers tablet and phone previews
Other Zoho services	Zoho Mail (notification workflows; planned integration)
Client hardware	Any computer able to run a current web browser, with internet access

3.5 Use Case Model

Figure 3.1 shows the use cases of the application. The vendor registers, uploads compliance documents and views the status of its own application. The Compliance Officer verifies documents, conducts compliance checks and requests corrections. The Procurement Manager records the approval decision. The Administrator manages users and permissions and views the reports and the dashboard. Registering a vendor includes a notification to the Compliance Officer, and recording a decision includes the update of the vendor status. Table 3.5 to Table 3.7 describe three key use cases in detail.

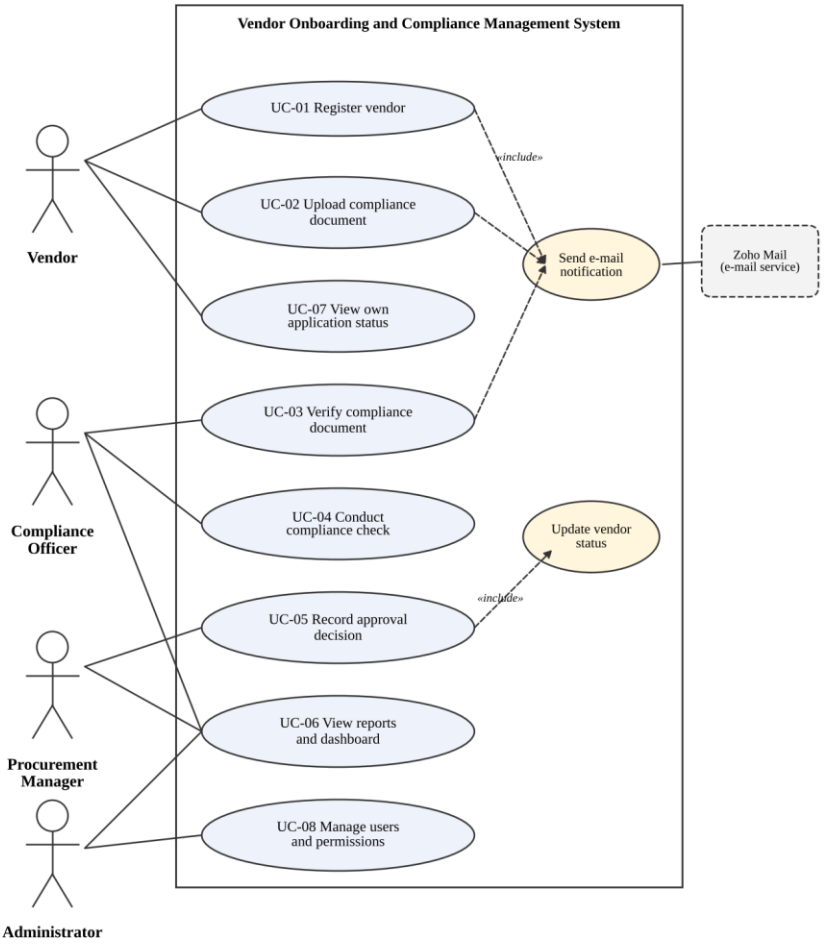


Figure 3.1: Use Case Diagram of the Vendor Onboarding and Compliance Management System

Table 3.5: Use Case Description – Register Vendor (UC-01)

Use Case ID	UC-01
Use Case Name	Register Vendor
Actor(s)	Vendor (primary); Administrator; Compliance Officer and e-mail service (secondary)
Pre-condition	The actor has access to the Vendor Registration form.
Main Flow	1. The actor opens the Vendor Registration form. 2. The actor enters the Company Name, Contact Person, Email, Mobile Number, Address, State, PIN Code, Vendor Type, PAN and GSTIN. 3. The actor clicks Submit. 4. The mandatory-field, mobile-number and PAN validations run. 5. The application generates a unique Vendor ID and sets the Vendor Status to Submitted. 6. The Compliance Officer receives an e-mail about the new registration.
Alternate Flow	If a mandatory field is empty or the mobile number or PAN is invalid, a validation message is shown and no vendor record is saved.
Post-condition	A vendor record with status Submitted appears in the All Vendors and Pending Applications reports.
Related FR	FR-01, FR-07

Table 3.6: Use Case Description – Verify Compliance Document (UC-03)

Use Case ID	UC-03
Use Case Name	Verify Compliance Document
Actor(s)	Compliance Officer (primary); Vendor and e-mail service (secondary)
Pre-condition	A compliance document with the status Pending exists; the actor has access to the Compliance Documents form.
Main Flow	1. The actor opens the document from the Document Verification report. 2. The actor reviews the uploaded file, the issue date and the expiry date. 3. The actor changes the Verification Status from Pending to Verified or Rejected and enters the Reviewer Remarks. 4. The actor saves the record. 5. The vendor receives an e-mail with the verification status and the remarks.
Alternate Flow	If the document is incomplete, the actor sets the status to Rejected and explains the reason in the remarks, and the vendor uploads a corrected document.
Post-condition	The document shows its new verification status and remarks, and the vendor has been notified.
Related FR	FR-02, FR-07

Table 3.7: Use Case Description – Record Approval Decision (UC-05)

Use Case ID	UC-05
Use Case Name	Record Approval Decision
Actor(s)	Procurement Manager (primary); Administrator; Vendor and e-mail service (secondary)
Pre-condition	The vendor’s documents and compliance checks have been reviewed; the actor has access to the Approval Records form.

Main Flow	1. The actor opens the Approval Records form. 2. The actor selects the Vendor and enters the Reviewer. 3. The actor selects the Decision (Approved, Rejected or Needs Correction), enters the Comments and the Decision Date. 4. The actor clicks Submit. 5. The Update Vendor Approval Status workflow sets the vendor status from the decision. 6. The vendor is notified of the decision.
Alternate Flow	If the decision is Needs Correction, the vendor updates the details or documents and the application returns to review.
Post-condition	The approval record appears in the All Approval Records report and in the Approval Pipeline board, and the vendor status has changed.
Related FR	FR-04, FR-07

CHAPTER 4

SYSTEM DESIGN

This chapter describes how the Vendor Onboarding and Compliance Management System is designed, before Chapter 5 shows how it was built. It covers the architecture, the data model, the form specifications, the vendor status process and workflows, the navigation and dashboard design, and the access-control design. The design described here is that of the application named Vendor Compliance System in the Zoho Creator console.

4.1 System Architecture

The application runs entirely inside Zoho Creator. It has no custom front end, no separate back end and no external database. Figure 4.1 shows the architecture. Vendors, compliance officers, procurement managers and administrators use the application through a web browser.

Presentation layer. Four forms (Vendor Registration, Compliance Documents, Compliance Checks and Approval Records), a set of list, calendar, Kanban and chart reports, and the Vendor Compliance Dashboard page.

Logic layer. Deluge workflows: form workflows that generate the Vendor ID, set initial statuses, update the vendor status after a decision and send notifications, and one scheduled workflow that runs daily for expiring documents.

Data layer. The records of the four forms, held by Creator and related by Vendor lookup fields: a compliance document, a compliance check and an approval record each refer to a vendor registration.

Notification channel. E-mail is sent through Zoho Mail notification workflows to the Compliance Officer and to vendors; this is the planned integration. No REST API or other external service is called.

Access control. Four roles – Administrator, Vendor, Compliance Officer and Procurement Manager – and permission profiles govern what each user can see and do, and vendors are restricted to their own records.

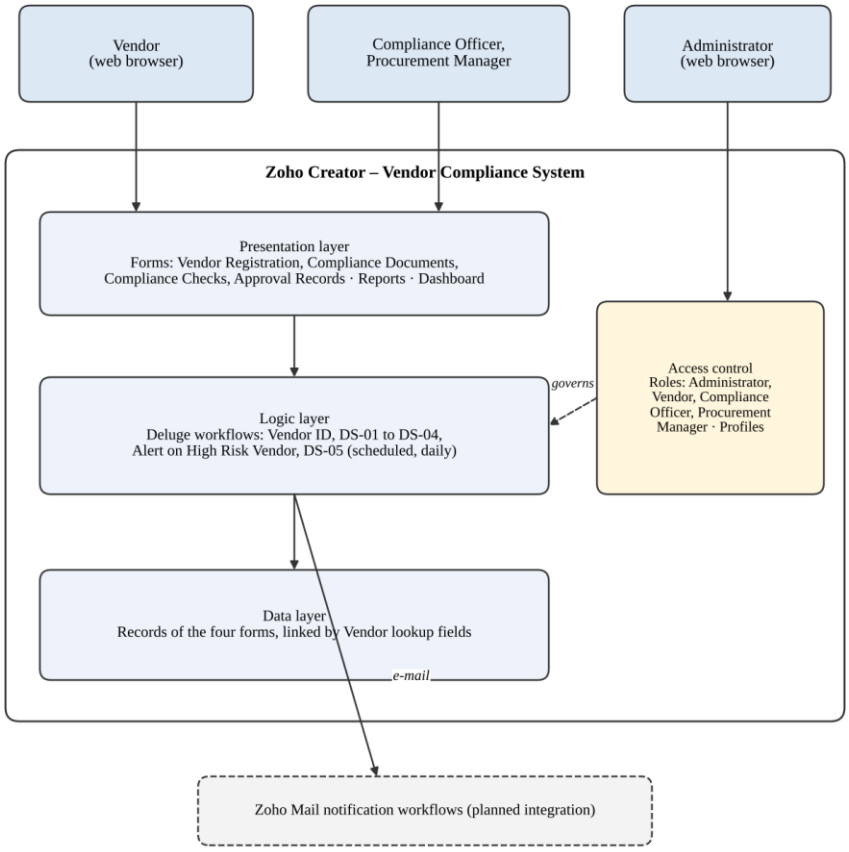


Figure 4.1: System Architecture of the Vendor Onboarding and Compliance Management System

4.2 Data Model

Zoho Creator stores data as records of forms. A form therefore plays the role that a table plays in a relational database, a field plays the role of a column, and a lookup field plays the role of a foreign key that points to a record of another form. The application uses four forms. Vendor Registration holds one record per vendor. Compliance Documents holds the documents of a vendor, Compliance Checks holds the checks made on a vendor and Approval Records holds the decisions made on a vendor; each carries a Vendor lookup, so one vendor can have many documents, many checks and many approval records, and each of these records belongs to a vendor. Figure 4.2 shows the entities and their relationships, and Table 4.1 summarises them.

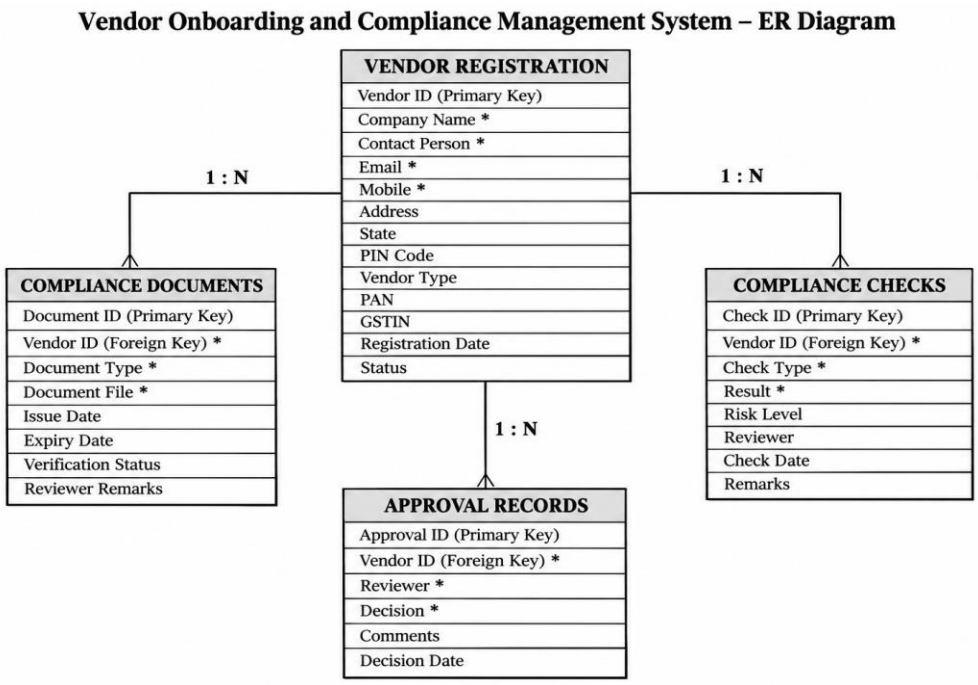


Figure 4.2: Entity–Relationship Diagram of the Vendor Onboarding and Compliance Management System (* marks a mandatory field)

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Using
Vendor Registration	Vendor ID, Company Name, Contact Person, Email, Mobile Number, Address, State, PIN Code, Vendor Type, PAN, GSTIN, Registration Date, Vendor Status	Compliance Documents; Compliance Checks; Approval Records	One-to-Many (to each)	Target of the Vendor lookup in the three other forms
Compliance Documents	Document ID, Vendor, Document Type, Document File, Issue Date, Expiry Date, Verification Status, Reviewer Remarks	Vendor Registration	Many-to-One	Vendor lookup
Compliance Checks	Check ID, Vendor, Check Type, Result, Risk Level, Reviewer, Check Date, Remarks	Vendor Registration	Many-to-One	Vendor lookup
Approval Records	Approval ID, Vendor, Reviewer, Decision, Comments, Decision Date	Vendor Registration	Many-to-One	Vendor lookup

Data consistency is supported in four ways.

- **Lookups.** The vendor on a document, a check or an approval record is chosen from existing vendor records, so mistyped company names cannot enter the data.
- **Generated identifier.** The Vendor ID is generated by a workflow, so each vendor has a unique identifier of the form VEN-nnnn (for example VEN-0027 in Figure 5.8).
- **Fixed value sets.** Vendor Status, Verification Status and Decision are drop-downs with the defined values only.

- **Validation on submission.** Mandatory fields, the mobile number and the PAN format are validated before a vendor record is saved (TC-02 to TC-04).

4.3 Form Design

Table 4.2 to Table 4.5 specify the fields of the four forms. Field names and mandatory flags were taken from the entity–relationship diagram, the form builder and the report columns; field types that are not visible in the screenshots are given as designed. Where scripts refer to a field, its link name is given in the Remarks column.

Table 4.2: Field Specification – Vendor Registration Form

Field Label	Field Type	Mandatory	Remarks
Vendor ID	Single Line (generated)	System	Primary key; generated as VEN-nnnn by a workflow
Company Name	Single Line	Yes	Link name company_name; shown in the Vendor lookups
Contact Person	Single Line	Yes	Contact person of the vendor
Email	Email	Yes	Link name email_address; used for vendor notifications
Mobile Number	Phone	Yes	Validated on submission (TC-03)
Address	Address	No	Vendor address
State	Drop Down	No	State of the vendor
PIN Code	Single Line	No	Postal Index Number
Vendor Type	Drop Down	No	Type of vendor
PAN	Single Line	No	Format validated on submission (TC-04)
GSTIN	Single Line	No	Goods and Services Tax Identification Number
Registration Date	Date	No	Link name submission_date, which DS-04 also sets to the current date
Vendor Status	Drop Down	No	Link name vendor_status; Submitted, Under Review, Approved, Rejected, Needs Correction
Verified By	Single Line (system)	No	Link name verified_by; set by DS-04 to the logged-in user

Table 4.3: Field Specification – Compliance Documents Form

Field Label	Field Type	Mandatory	Remarks
Vendor	Lookup (Vendor Registration)	Yes	Link name vendor; displays Company Name; display type Multi Select in Figure 5.2
Document Type	Drop Down	Yes	Values seen: Incorporation Certificate, GST Certificate, Other
Document File	File Upload	Yes	Uploaded file, for example ISO_9001_Cert_2026.pdf
Issue Date	Date	No	Date of issue of the document
Expiry Date	Date	No	Link name expiry_date; read by the scheduled workflow DS-05

Field Label	Field Type	Mandatory	Remarks
Verification Status	Drop Down	No	Link name verification_status; Pending, Verified, Rejected
Reviewer Remarks	Multi Line	No	Link name reviewer_comments; remarks of the Compliance Officer

Table 4.4: Field Specification – Compliance Checks Form

Field Label	Field Type	Mandatory	Remarks
Vendor	Lookup (Vendor Registration)	Yes	Vendor to which the check applies
Check Type	Drop Down	Yes	Values seen: GST Verification, PAN Verification, Document Review, Business Details
Result	Drop Down	Yes	Values seen: Pass, Fail, Needs Review
Risk Level	Drop Down	No	Values seen: Low, Medium; High is recorded for high-risk checks
Reviewer	Single Line	No	Name of the reviewer
Check Date	Date	No	Date of the check
Remarks	Multi Line	No	Remarks of the reviewer

Table 4.5: Field Specification – Approval Records Form

Field Label	Field Type	Mandatory	Remarks
Vendor	Lookup (Vendor Registration)	Yes	Vendor to which the decision applies
Reviewer	Single Line	Yes	Name of the reviewer who decides
Decision	Drop Down	Yes	Link name decision; Approved, Rejected, Needs Correction
Comments	Multi Line	No	Comments to the vendor
Decision Date	Date	No	Date of the decision

4.4 Process and Workflow Design

Vendor status process. A vendor is registered with the status Submitted. The Compliance Officer reviews the application and its documents and moves the status to Under Review. If the documents or checks are incomplete, the status becomes Needs Correction and the vendor updates the details or documents, which returns the application to review. If all requirements pass, the Procurement Manager decides: Approved or Rejected. A compliance failure leads to Rejected. Figure 4.3 shows the end-to-end system workflow, Figure 4.4 shows the onboarding activity and Figure 4.5 shows the status diagram; Table 4.6 lists the vendor status values.

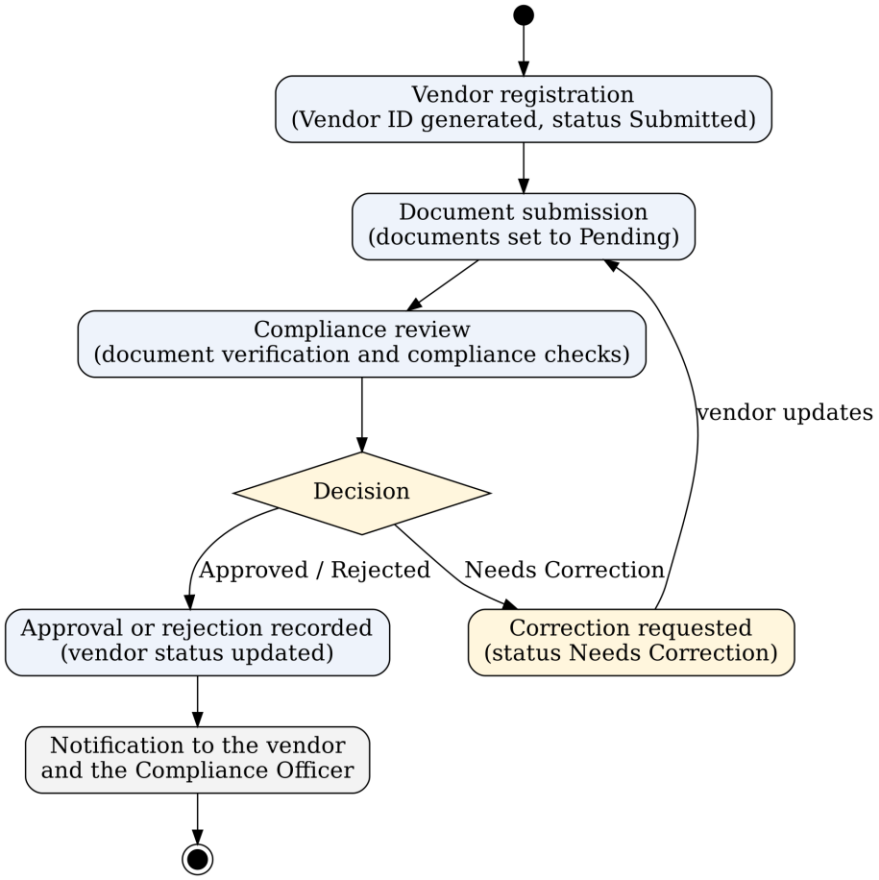


Figure 4.3: System Workflow of Vendor Onboarding and Compliance Review

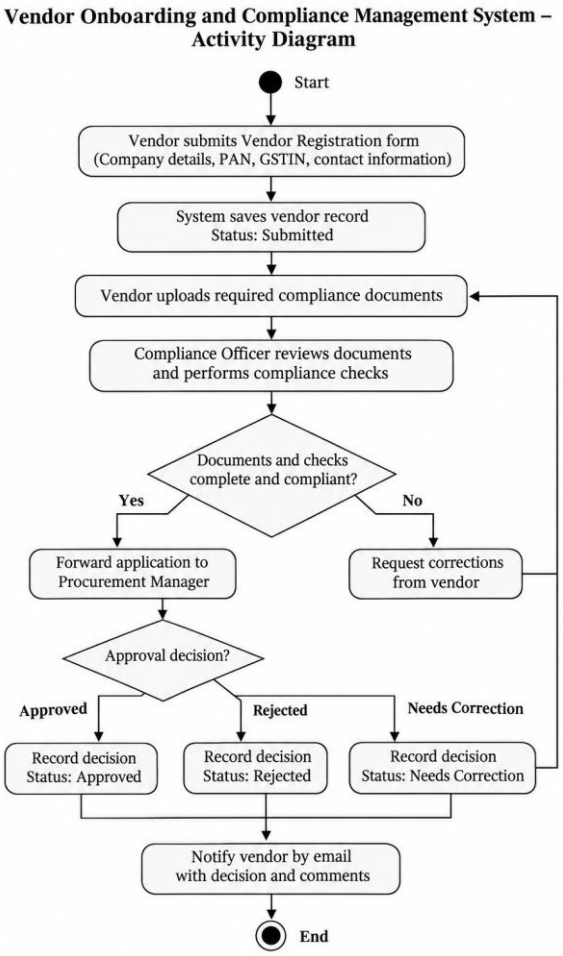


Figure 4.4: Activity Diagram of the End-to-End Vendor Onboarding Process

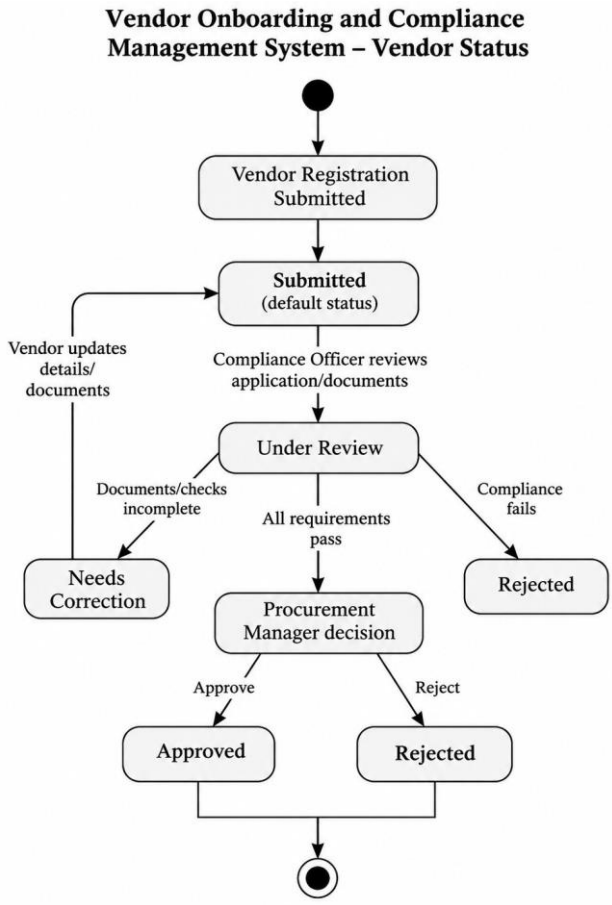


Figure 4.5: Vendor Status Diagram

Table 4.6: Vendor Status Values and Transitions

Status	Entered When	Changed By	Where It Appears
Submitted	The vendor submits the registration; set automatically after registration	Workflow DS-01 (automated)	All Vendors and Pending Applications reports
Under Review	The Compliance Officer starts reviewing the application and documents	Compliance Officer (manual edit)	Pending Applications report; dashboard
Needs Correction	Documents or checks are incomplete, or the decision is Needs Correction	Compliance Officer; workflow DS-04 for the decision	Approval Pipeline board; All Approval Records report
Approved	The Procurement Manager records the decision Approved	Workflow DS-04 (automated)	All Vendors report; Approval Pipeline board
Rejected	Compliance fails, or the Procurement Manager records the decision Rejected	Workflow DS-04 (automated)	All Vendors report; Approval Pipeline board

Workflows. The workflows are listed in Table 5.6. Four form workflows, DS-01 to DS-04, set statuses and send notifications; a workflow generates the Vendor ID; a form workflow alerts on a high-risk vendor; and DS-05 is the daily scheduled workflow. The event decides when a workflow runs: Successful form submission runs after the record is saved, On update of a field runs when that field is edited, and a scheduled workflow runs at a set time.

Notification design. Five notifications leave the application: a new vendor registration and a new document upload are notified to the Compliance Officer; a document verification result, an approval or rejection decision and a document expiry reminder are notified to the vendor. Each is sent as an e-mail through the Zoho Mail notification workflows, which are the planned integration.

Document expiry logic. The scheduled workflow selects every compliance document whose expiry date is not empty, is not earlier than today and is not later than 30 days from today, and e-mails the vendor concerned. The rule is

Reminder is sent when Expiry Date ≥ today and Expiry Date ≤ today + 30 days

so a document that expires on the thirtieth day from today is included and one that has already expired is not. For example, on 2 November 2026 a document expiring on 25 November 2026 would trigger a reminder, and one expiring on 1 January 2027 would not. The defect BUG-05 (Section 6.4) concerned the correct checking of this window.

The design has stated limitations. The reminder is sent every day while the document stays inside the 30-day window, so a vendor may receive more than one reminder for the same document. A document that has already expired is not reminded. The vendor status after a decision is set by the workflow and cannot be moved forward by the vendor.

4.5 User Interface and Navigation Design

Figure 4.6 shows the navigation structure of the application. The side menu has the Dashboard and four sections, each pairing a data-entry form with its reports. The Vendor Registrations section holds the Vendor Registration form and the All Vendors report. The Compliance Documents section holds the Compliance Documents form, the Document Verification report and the Expiring Documents calendar. The Compliance Checks section holds the Compliance Checks form and the Compliance Overview chart. The Approval Records section holds the Approval Records form and the Approval Pipeline Kanban board. The Dashboard holds the KPI summary, the recent applications and the quick actions.

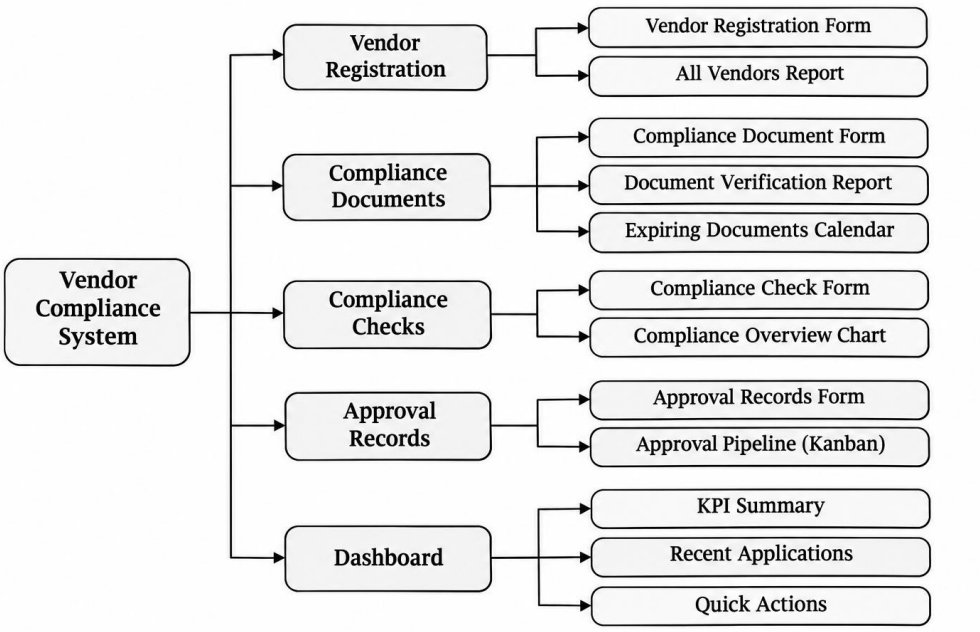


Figure 4.6: Navigation Structure of the Vendor Onboarding and Compliance Management System

Dashboard. The Vendor Compliance Dashboard is the landing page for internal users. It combines KPI tiles for total vendors, pending approvals, approved vendors and documents expiring within 30 days, a compliance chart, a list of recent applications and a Quick Links panel with buttons to add a vendor registration, a compliance document, a compliance check and an approval record. A person can therefore see the state of the onboarding process on one screen and open a form in one click.

Forms and reports. Each form is a single page of labelled fields, so a user fills it from top to bottom. List reports show records in rows with a search icon and an add button, the Approval Pipeline board shows approval records as cards grouped by decision, and the interface uses the platform's blue side menu with a red highlight for the selected item and can be previewed in desktop, tablet and phone layouts.

4.6 Access Control Design

Access is designed on three levels. Users sign in with their accounts. Four roles – Administrator, Vendor, Compliance Officer and Procurement Manager – describe the kinds of user. Permission profiles define what each kind of user may do with each form, report and page, and a record-visibility rule restricts a vendor to its own records. Vendors must not reach the internal reports or the approval functions.

Role design. Table 4.7 records the intent of each role as written in the project plan. How the permission profiles of the application are set is shown in Chapter 5.

Table 4.7: Permission Profiles and Their Design Intent

Role	Intended Access	Intended Restriction
Administrator	Manage application configuration, users, permissions and records	None
Vendor	Register, maintain permitted profile details, upload documents and view own status	Own records only; no internal reports and no approval functions
Compliance Officer	Review applications, verify documents, conduct compliance checks and request corrections	No final approval decision
Procurement Manager	Review compliance results and approve, reject or request corrections	No administration of users or application settings

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator account under the name Vendor Compliance System; its link name is vendor-compliance-system and the application address is https://creatorapp.zoho.in/nithiyasreenc/vendor-compliance-system. The components are organised in the application menu as the Dashboard, the Compliance Documents section, the Approval Records section (with the entries Approval Records, All Approval Records and Decisions), the Compliance Checks section and the Vendor Registrations section (Figure 5.1). Table 5.1 summarises the set-up.

The application was developed in the Development environment on the trial edition; the application header shows “Trial expires in 14 days” (Figure 5.1). Publication across environments is not evidenced in the supplied screenshots (Section 7.1).

Table 5.1: Application Set-up Summary

Attribute	Details
Application name	Vendor Compliance System
Link name	vendor-compliance-system
Edition	Trial edition (14 days remaining when the dashboard was captured)
Menu	Dashboard; Compliance Documents; Approval Records (Approval Records, All Approval Records, Decisions); Compliance Checks; Vendor Registrations
Forms	Vendor Registration, Compliance Documents, Compliance Checks, Approval Records
Environment	Development
Logged-in user	Nithiya Sree (application owner)

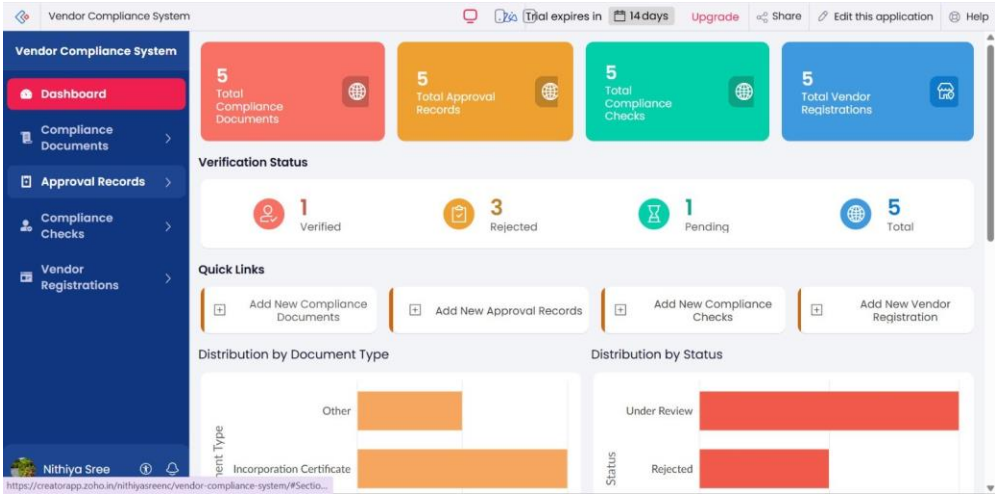


Figure 5.1: Vendor Compliance Dashboard

5.2 Forms and Data Capture

The application stores its data in four forms: Vendor Registration, Compliance Documents, Compliance Checks and Approval Records. Table 5.2 summarises the forms; the full field specification is given in Table 4.2 to Table 4.5 of Chapter 4.

Table 5.2: Forms Implemented

Form	Sections	Mandatory Fields	Special Features
Vendor Registration	Single page: identity (Vendor ID, Company Name, Contact Person), contact (Email, Mobile Number, Address, State, PIN Code), registration (Vendor Type, PAN, GSTIN, Registration Date, Vendor Status)	Company Name, Contact Person, Email, Mobile Number	Generated Vendor ID; status set to Submitted; PAN and mobile validation; workflows DS-01 and DS-04 act on it; target of the Vendor lookups
Compliance Documents	Vendor, Document Type, Document File, Issue Date, Expiry Date, Verification Status, Reviewer Remarks	Vendor, Document Type, Document File	Vendor lookup; File Upload field; status set to Pending; workflows DS-02 and DS-03; scheduled workflow DS-05
Compliance Checks	Vendor, Check Type, Result, Risk Level, Reviewer, Check Date, Remarks	Vendor, Check Type, Result	Vendor lookup; Alert on High Risk Vendor workflow
Approval Records	Vendor, Reviewer, Decision, Comments, Decision Date	Vendor, Reviewer, Decision	Vendor lookup; workflow DS-04 updates the vendor status

Vendor Registration form. It captures the vendor identity and registration information: Vendor ID, Company Name, Contact Person, Email, Mobile Number, Address, State, PIN Code, Vendor Type, PAN, GSTIN, Registration Date and Vendor Status. The Vendor ID is generated by a workflow, and the Vendor Status is set to Submitted after registration. The form was not captured in a screenshot, so its layout is described from the specification.

Compliance Documents form. Figure 5.2 shows the form in the form builder. The fields are Vendor, Document Type, Document File, Issue Date, Expiry Date, Verification Status and Reviewer Remarks. Vendor is a lookup field to the Vendor Registration form: the Field Properties panel shows the field name Vendor, the link name vendor, the display field Company Name from Vendor Registration and the display type Multi Select. Because the display type is Multi Select, a document record may be linked to more than one vendor in the sample data (Figure 5.3); setting the display type to Drop Down would enforce one vendor per record, as in the one-to-many design of Figure 4.2.

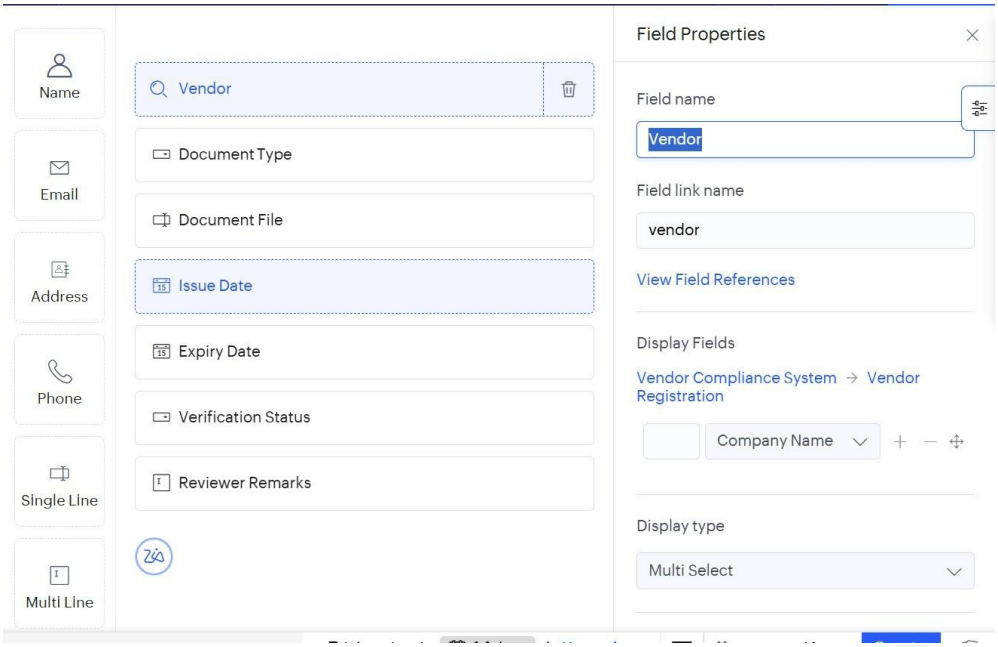


Figure 5.2: Compliance Documents Form in the Form Builder with the Vendor Lookup Properties

Compliance Checks form. It records the Vendor, Check Type, Result, Risk Level, Reviewer, Check Date and Remarks of a compliance check. The sample records show the check types GST Verification, PAN Verification, Document Review and Business Details (Figure 5.4).

Approval Records form. It records the Vendor, the Reviewer, the Decision (Approved, Rejected or Needs Correction), the Comments and the Decision Date (Figure 5.5). The Vendor lookup in the Compliance Checks and Approval Records forms ties every check and decision to a vendor, which gives the one-to-many relationships shown in Figure 4.2.

5.3 Reports

The application provides six reports, which Table 5.3 summarises. The Document Verification report is shown in Figure 5.3: it lists the Vendor, Document Type, Document File, Issue Date and Expiry Date of the five stored documents, and the document type cells are colour-coded. The All Compliance Checks report (Figure 5.4) lists the Vendor, Check Type, Result, Risk Level, Reviewer, Check Date and Remarks of five checks, with the check type colour-coded. The All Approval Records report (Figure 5.5) lists the Vendor, Reviewer, Decision, Comments and Decision Date of five records and shows “Showing 5 of 5”; the decision is colour-coded. The Approval Pipeline Kanban report (Figure 5.6), opened from the Decisions entry of the menu, groups the approval records as cards in the columns Approved, Needs Correction and Rejected: Sarah Connor and Emily White are in Approved, Lisa Anderson is in Needs Correction, and John Smith and Michael Brown are in Rejected.

Table 5.3: Reports Implemented

Report Name	Menu Section	Type	Source Form	Evidence / Purpose
All Vendors	Vendor Registrations	List	Vendor Registration	Output not shown; lists all vendor records
Pending Applications	Vendor Registrations	List	Vendor Registration	Output not shown; lists vendors that are not yet decided
Document Verification	Compliance Documents	List	Compliance Documents	Figure 5.3; five records
Expiring Documents	Compliance Documents	Calendar	Compliance Documents	Output not shown; documents by expiry date
Approval Pipeline	Approval Records	Kanban	Approval Records	Figure 5.6; columns by decision
Compliance Overview	Compliance Checks	Chart	Compliance Checks	Output not shown; the dashboard charts are shown in Figure 5.1

The report entries seen in the menu in the screenshots are All Compliance Documents, All Compliance Checks and All Approval Records, together with the Decisions board; they are the list and board views of the reports named in Table 5.3. The sample comments of the approval records are test data and do not always agree with the decision shown in the same row.



Figure 5.3: Document Verification Report (All Compliance Documents)

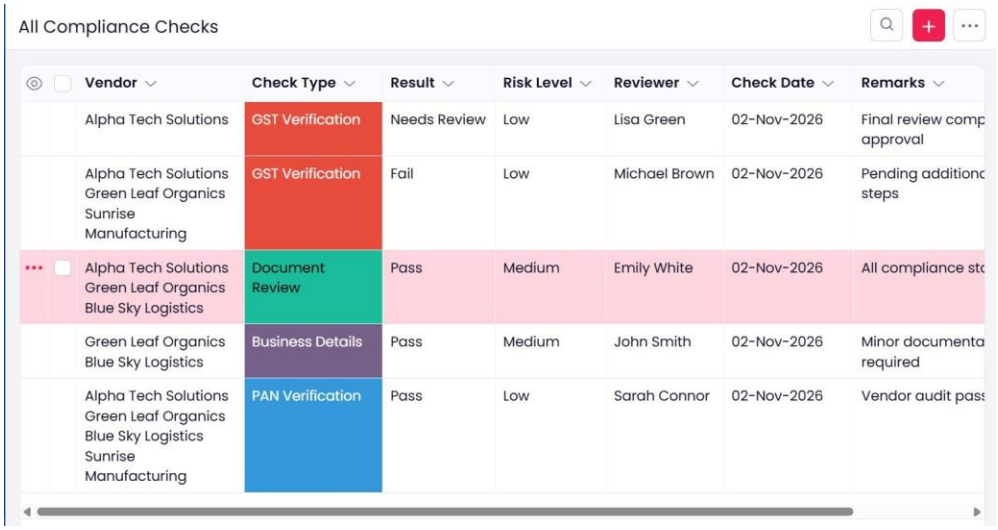


Figure 5.4: All Compliance Checks Report



Figure 5.5: All Approval Records Report

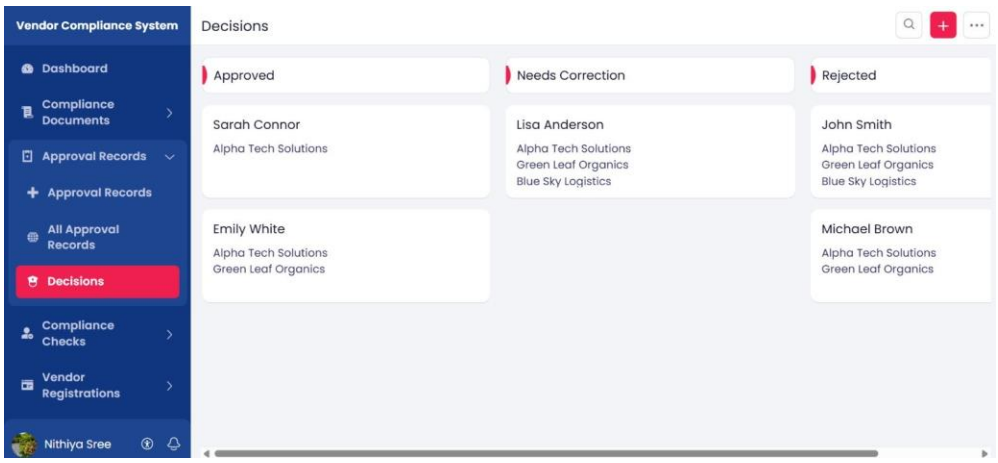


Figure 5.6: Approval Pipeline Kanban Report (Decisions Board)

5.4 Pages and Dashboards

The Vendor Compliance Dashboard (Figure 5.1) is the first entry in the menu and is built with the page builder. From top to bottom it contains four KPI tiles, a Verification Status

block, a Quick Links panel and two charts. The KPI tiles show 5 Total Compliance Documents, 5 Total Approval Records, 5 Total Compliance Checks and 5 Total Vendor Registrations. The Verification Status block shows 1 Verified, 3 Rejected, 1 Pending and 5 Total. The Quick Links panel has the buttons Add New Compliance Documents, Add New Approval Records, Add New Compliance Checks and Add New Vendor Registration. The charts are Distribution by Document Type and Distribution by Status. Table 5.4 lists the components designed for the dashboard and the values visible in the screenshot; the components that are not visible in the captured part of the page are marked as such.

The values agree with the reports. The five documents in Figure 5.3, the five checks in Figure 5.4 and the five approval records in Figure 5.5 equal the tile values, and the Verification Status total of 5 equals the number of documents.

Table 5.4: Components of the Dashboard

Component	Type	Data Source	Value Observed
Total Vendors (Total Vendor Registrations)	KPI tile	Vendor Registration	5
Total Compliance Documents	KPI tile	Compliance Documents	5
Total Compliance Checks	KPI tile	Compliance Checks	5
Total Approval Records	KPI tile	Approval Records	5
Verification Status	Status block	Compliance Documents	Verified 1, Rejected 3, Pending 1, Total 5
Pending Approvals KPI	KPI tile	Vendor Registration	Not visible in the captured part of the page
Approved Vendors KPI	KPI tile	Vendor Registration	Not visible in the captured part of the page
Documents Expiring Within 30 Days KPI	KPI tile	Compliance Documents	Not visible in the captured part of the page
Distribution by Document Type	Bar chart	Compliance Documents	Bars for Other and Incorporation Certificate and further types below the visible area
Distribution by Status	Bar chart	Vendor Registration	Bars for Under Review and Rejected and further statuses below the visible area
Recent Applications	List	Vendor Registration	Not visible in the captured part of the page
Quick Links	Four buttons	Forms	Add New Compliance Documents, Approval Records, Compliance Checks and Vendor Registration

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator; it is used to write form-event logic,

functions and integrations, for example to create or update records [2]. All workflows of the application run Deluge actions, and Table 5.5 lists the scripts. They are reproduced in full in Appendix C.

Table 5.5: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose
DS-01	Vendor Registration → Created → Successful form submission (workflow "Notify on New Vendor Registration")	Sets the vendor status to Submitted and e-mails the Compliance Officer
DS-02	Compliance Documents → Created → Successful form submission (workflow "Notify on Document Upload")	Sets the verification status to Pending and e-mails the Compliance Officer
DS-03	Compliance Documents → Edited → On update of Verification Status (workflow "Notify Vendor on Document Verification")	E-mails the vendor the verification result and the reviewer remarks
DS-04	Approval Records → Created → Successful form submission (workflow "Update Vendor Approval Status")	Updates the vendor status from the decision and records the reviewer and date
DS-05	Compliance Documents → Scheduled workflow → Daily (workflow "Notify on Expiring Documents")	E-mails vendors about documents expiring within 30 days

Update Vendor Approval Status (DS-04). After an approval record is saved, the script fetches the vendor record whose ID equals the vendor on the approval record. If the decision is Approved, Rejected or Needs Correction, it sets the vendor status to the same value. It then stores the logged-in user in the field verified_by and the current date in the field submission_date. Code 5.1 reproduces the script.

Code 5.1: DS-04 Update Vendor Approval Status (Approval Records → Created → Successful form submission)

```
vendorRecord = vendor_registration[ID == input.vendor];

if(input.decision == "Approved")
{
    vendorRecord.vendor_status = "Approved";
}
else if(input.decision == "Rejected")
{
    vendorRecord.vendor_status = "Rejected";
}
else if(input.decision == "Needs Correction")
{
    vendorRecord.vendor_status = "Needs Correction";
}

vendorRecord.verified_by = zoho.loginuser;
vendorRecord.submission_date = zoho.currentdate;
```

Notify on Expiring Documents (DS-05). The scheduled workflow loops over the compliance documents. For each document whose expiry date is not empty, is not earlier than the current date and is not later than 30 days from the current date, it fetches the vendor record and, if the vendor's e-mail address is not empty, sends the reminder e-mail. Code 5.2 reproduces the script.

Code 5.2: DS-05 Notify on Expiring Documents (Compliance Documents → Scheduled workflow → Daily)

```
for each doc in vendor_document
{
    if(doc.expiry_date != null
       && doc.expiry_date >= zoho.currentdate
       && doc.expiry_date <= zoho.currentdate.addDay(30))
    {
        vendorRecord = vendor_registration[ID == doc.vendor];

        if(vendorRecord.email_address != null
           && vendorRecord.email_address != "")
        {
            sendmail
            [
                from : zoho.adminuserid
                to : vendorRecord.email_address
                subject : "Compliance Document Expiry Reminder"
                message : "Dear " + vendorRecord.company_name + ",\n\n"
                        + "Your " + doc.document_type
                        + " is due to expire on " + doc.expiry_date + ".\n"
                        + "Please upload an updated document before the expiry date."
            ];
        }
    }
}
```

The other scripts have the same pattern: DS-01 and DS-02 set an initial status and send an e-mail to the Compliance Officer, and DS-03 sends an e-mail to the vendor only when the vendor's e-mail address is not empty. Standalone functions are not used, and the only loop is the one in DS-05.

5.6 Workflows and Automation

The workflows are listed in Table 5.6 and the workflow page is shown in Figure 5.7. The page lists Compliance Checks and Vendor Registration under Forms with workflows and Approval Records and Compliance Documents under Forms without workflows. For Compliance Checks it shows the workflow Alert on High Risk Vendor, triggered on Created or Edited → Successful form submission, with 1 action, the status Enabled and the creation date 02-Nov-2026. The workflows DS-02 to DS-04 and the daily schedule DS-05 are specified in Appendix C and in Table 5.5; they appear in the capture only as far as the workflow page shows.

Table 5.6: Workflows Implemented

Workflow Name	Form	Event / Trigger	Purpose
Generate Vendor ID	Vendor Registration	Created	Generates the unique Vendor ID (VEN-nnnn)
Notify on New Vendor Registration (DS-01)	Vendor Registration	Created → Successful form submission	Sets the status to Submitted; notifies the Compliance Officer
Notify on Document Upload (DS-02)	Compliance Documents	Created → Successful form submission	Sets the document to Pending; notifies the Compliance Officer
Notify Vendor on Document Verification (DS-03)	Compliance Documents	Edited → On update of Verification Status	Notifies the vendor of the verification result

Workflow Name	Form	Event / Trigger	Purpose
Update Vendor Approval Status (DS-04)	Approval Records	Created → Successful form submission	Updates the vendor status from the decision
Alert on High Risk Vendor	Compliance Checks	Created or Edited → Successful form submission (Enabled, 1 action)	Alerts when a high-risk check is recorded (Figure 5.7)
Notify on Expiring Documents (DS-05)	Compliance Documents	Scheduled → Daily	Reminds vendors of documents expiring within 30 days

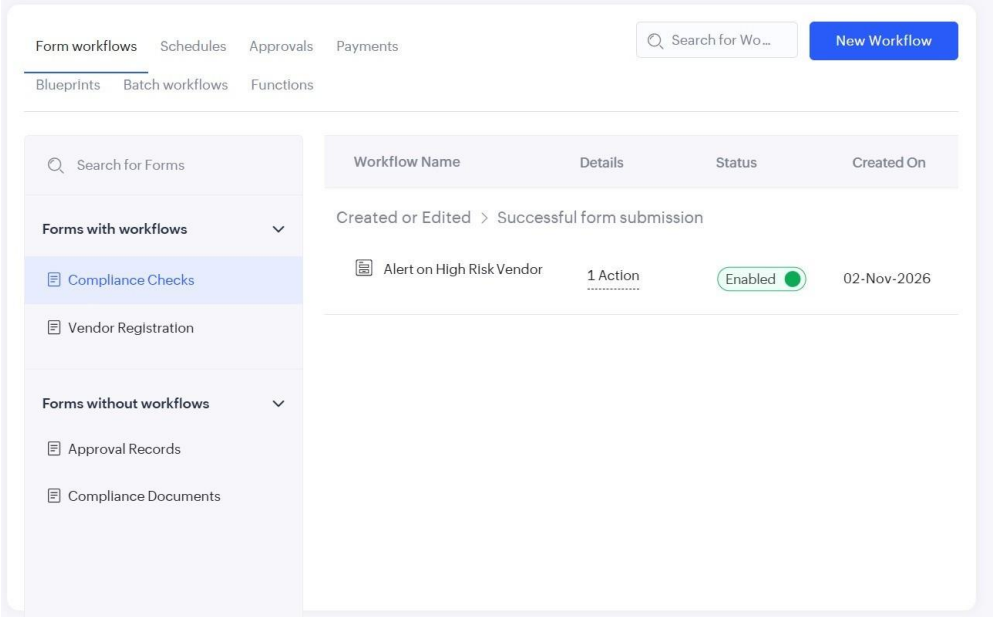


Figure 5.7: Form Workflows Page Showing the Alert on High Risk Vendor Workflow of Compliance Checks

Status process. The vendor status moves through Submitted, Under Review, Approved, Rejected and Needs Correction (Figure 4.5). The Approval Records form with the decision-based status update of DS-04 forms the approval step; a separate approval process and a Blueprint are planned and are not configured as separate components. No batch workflow is used.

5.7 Integrations

Table 5.7 summarises the integrations. **Zoho Mail.** The notifications are produced by Zoho Mail notification workflows, which are the planned integration; the Deluge scripts send mail with the sendmail statement from the administrator's address. Figure 5.8 shows the format of the notification that the Compliance Officer receives for a new vendor registration: the subject "New Vendor Application Submitted – VEN-0027", the Vendor ID, the company name, the contact person, the e-mail address, the mobile number and the submission date, and a request to review the application in the Vendor Compliance System. The screenshot is

an example of the notification format; it does not by itself evidence a live Zoho Mail integration, which is recorded as Partial in Appendix A. **REST API.** No REST API call, webhook or Postman collection is used.

Table 5.7: Integrations and Notifications

Notification	Recipient	Trigger	Source Script	Purpose
New vendor registration	Compliance Officer	Vendor Registration → Successful form submission	DS-01	Request a review of the new application (Figure 5.8)
New document upload	Compliance Officer	Compliance Documents → Successful form submission	DS-02	Request verification of the document
Document verification result	Vendor	Compliance Documents → On update of Verification Status	DS-03	Tell the vendor the status and the reviewer remarks
Approval or rejection decision	Vendor	Approval Records → Successful form submission	Approval workflow (notification script not reproduced)	Tell the vendor the decision and the comments
Document expiry reminder	Vendor	Scheduled → Daily	DS-05	Remind the vendor to upload an updated document

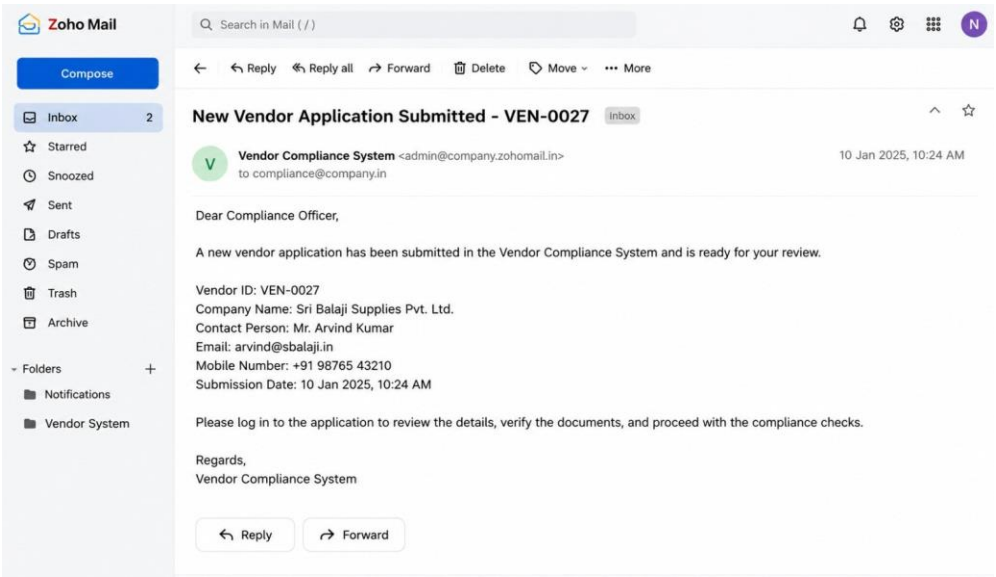


Figure 5.8: Example of the New Vendor Application Notification E-mail (Zoho Mail)

5.8 User Management and Role-Based Access

Access control is built from roles, permission profiles and record-visibility rules. Figure 5.9 shows the Permissions tab of Settings → User Permissions. It lists the permission profiles Approval Authority (make final decisions on vendor registration status including approval or rejection), Compliance Auditor (verify PAN and GSTIN details against government

databases), Risk Assessment Manager (analyse aggregated compliance check results to determine overall vendor risk) and the default Read profile (read permission for all components). The four roles of Table 3.1 are the roles of the application design. Table 5.8 summarises the access rules that apply to them: vendors reach only their own records, and the internal reports and the approval functions are not accessible to vendors.

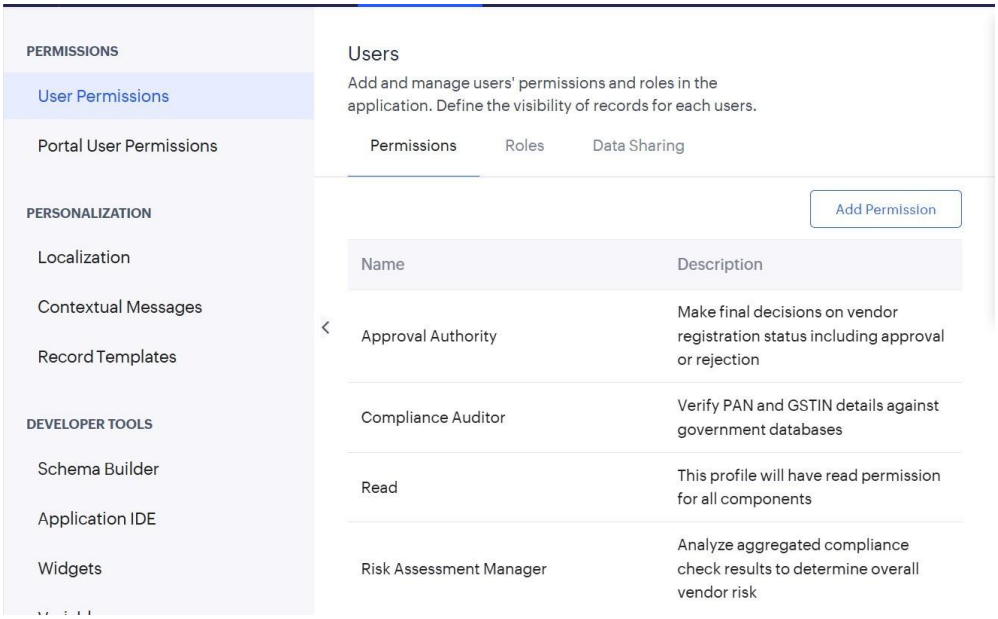


Figure 5.9: Permission Profiles of the Application (Settings → Users → Permissions)

Table 5.8: Role-Based Access Rules

Role	Forms	Reports and Dashboard	Record Access
Administrator	All four forms	All reports and the dashboard	All records
Vendor	Vendor Registration; Compliance Documents (upload)	Own status only	Own records only
Compliance Officer	Compliance Documents; Compliance Checks; Vendor Registration (review)	All internal reports and the dashboard	All vendor records for review
Procurement Manager	Approval Records	Approval Pipeline, compliance results and the dashboard	All vendor records for decision

The assignment of individual users to the four roles, a sign-in test of each role and the record-level criteria that restrict a vendor to its own records are not shown in the screenshots; they are recorded as Partial in Appendix A, and the defect BUG-06 (Section 6.4) concerned the correction of the vendor record visibility.

5.9 AI and Analytics Features

No generative or predictive AI feature is part of the application. The Zia assistant icon is visible in the builder screens, but no Zia-generated component or AI prediction is used, and

AI-based risk prediction is outside the scope of the project.

Analytics are provided by the dashboard (Figure 5.1). Four KPI tiles and a Verification Status block give live counts, and the charts Distribution by Document Type and Distribution by Status compare the documents and the vendor statuses. Zoho Analytics is not used. Further charts, such as vendors by risk level, are listed as future enhancements in Section 9.2.

5.10 Pro-Code Extension

No pro-code extension was used. The application does not use Zoho Catalyst, custom widgets or an external service written in another language. All logic is implemented with Creator workflows and Deluge, which was sufficient for the status, notification and expiry-reminder requirements.

5.11 Mobile and Portal Access

Vendor portal. Vendors are planned to reach their own records through a vendor portal that shows only their registration, their documents and their status. The portal configuration was not captured in a screenshot, so it is recorded as Partial in Appendix A.

Mobile access. The application header offers desktop, tablet and phone preview icons (Figure 5.1). The platform renders applications responsively, but no screenshot of the application on a phone or in the Zoho Creator mobile app was supplied, and no mobile-specific configuration is claimed.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was functional and manual. The application was exercised through its live interface in the Development environment of Zoho Creator. Each feature described in Chapters 4 and 5 was checked by entering records, observing the reports and the dashboard, reading the notification e-mails produced by the workflows, and opening the Deluge scripts in the editor. Sixteen functional test cases were defined against the functional requirements FR-01 to FR-07 and were run; the result of each is recorded in Table 6.1.

Checking was done at four levels:

- **Unit level.** Single forms and fields (mandatory flags, mobile number and PAN validation, default statuses) and single scripts (reading the code in the editor).
- **Integration level.** Workflow to e-mail, approval record to vendor status, and the dashboard figures compared with the contents of the reports.
- **Security level.** Roles, permission profiles and vendor record visibility compared with their intended design.
- **Regression level.** Re-running the affected test cases after each defect in Table 6.4 was corrected.

6.2 Test Cases

Table 6.1 lists the test cases. Status values are Pass, Fail and Not Run.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit vendor registration	Enter required vendor details and submit	Vendor record saved with unique ID	Vendor record saved with generated ID	Pass
TC-02	FR-01	Validate mandatory vendor fields	Leave Company Name empty and submit	Validation message displayed	Validation message displayed	Pass
TC-03	FR-01	Validate mobile number	Enter invalid mobile number	Invalid number rejected	Invalid number rejected	Pass
TC-04	FR-01	Validate PAN format	Enter invalid PAN	Validation error displayed	Invalid PAN rejected	Pass
TC-05	FR-02	Upload compliance document	Select vendor, document type, file, issue date and expiry date	Document saved with Pending status	Document saved as Pending	Pass
TC-06	FR-02	Verify uploaded document	Change Pending to Verified	Verification status updated	Status updated to Verified	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-07	FR-02	Reject a compliance document	Set status to Rejected and enter remarks	Rejection and remarks saved	Rejection and remarks saved	Pass
TC-08	FR-03	Create compliance check	Enter vendor, check type, result, risk, reviewer and remarks	Compliance check saved	Compliance check saved	Pass
TC-09	FR-03	Review high-risk check	Create a High risk compliance check	High-risk check recorded	High-risk check recorded	Pass
TC-10	FR-04	Create approval record	Enter vendor, reviewer, decision and comments	Approval record saved	Approval record saved	Pass
TC-11	FR-04	Approve vendor	Record Approved decision	Vendor status becomes Approved	Status updated to Approved	Pass
TC-12	FR-04	Reject vendor	Record Rejected decision	Vendor status becomes Rejected	Status updated to Rejected	Pass
TC-13	FR-05	View All Vendors report	Open All Vendors	Vendor records displayed	Vendor records displayed	Pass
TC-14	FR-05	View Pending Applications	Open Pending Applications	Pending applications displayed	Pending applications displayed	Pass
TC-15	FR-06	View dashboard	Open Vendor Compliance Dashboard	KPIs, charts and quick actions displayed	Dashboard displayed correctly	Pass
TC-16	FR-07	Verify email notification	Update document status or approval decision	Relevant email sent to vendor	Notification email received	Pass

Table 6.2 summarises the results. No test case failed and none was left unrun. The defects found while the application was being built, which are recorded in Table 6.3 and Table 6.4, were corrected before the test cases were run to completion.

Table 6.2: Summary of Test Results

Result	Count	Test Cases	Comment
Pass	16	TC-01 to TC-16	All documented cases passed
Fail	0	—	No failed cases
Not Run	0	—	All documented cases completed

6.3 Debugging Deluge Scripts

Three debugging aids are available in the Deluge editor: the Execute button, the info statement that writes a value to the log, and the output that each workflow leaves behind (a changed status or an e-mail). Reading the scripts and comparing the records they change gave the findings below.

Update Vendor Approval Status. The workflow runs after an approval record is saved. The script fetches the vendor with the ID stored in the vendor lookup of the approval record, compares the decision with the three values Approved, Rejected and Needs Correction and copies the matching value to the vendor status. Reading the code gives two points to check.

First, if the decision field is empty or holds another value, no status is set, so the decision must be one of the three values. Second, the script writes the logged-in user to verified_by and the current date to submission_date, so the registration date field is overwritten at each decision (Section 4.3).

Notify Vendor on Document Verification. The workflow runs when the verification status is edited. The script fetches the vendor of the document and sends the e-mail only when the vendor's e-mail address is neither null nor empty; a vendor without an e-mail address therefore receives nothing, which is the cause that was checked in BUG-04.

Notify on Expiring Documents. The scheduled workflow loops over every document and tests the expiry date against the current date and the current date plus 30 days. The test of the lower bound keeps documents that have already expired out of the reminders, and the loop visits every document on every run, which is noted as an optimisation opportunity in Table 7.2.

Table 6.3 lists the symptoms that were traced in this way and what each trace showed.

Table 6.3: Debugging Observations

No.	Symptom	Technique Used	Finding	Defect
1	Vendor ID was not generated	Checked registration workflow and saved record	Vendor ID generation logic was missing	BUG-01
2	Document saved without Pending status	Compared submitted document and stored status	Initial verification status was not set	BUG-02
3	Vendor status did not update after approval	Compared Approval Records and Vendor Registration	Approval workflow did not update related vendor	BUG-03
4	Vendor did not receive rejection email	Checked workflow and vendor email field	Rejection notification was incorrectly configured	BUG-04
5	Expiring documents were not included in reminders	Reviewed scheduled workflow and date condition	Expiry reminder window was not checked correctly	BUG-05
6	Vendor could access another vendor's records	Tested vendor access and reviewed permissions	Record visibility restrictions needed correction	BUG-06

6.4 Defect Log

Table 6.4 records the six defects of Table 6.3. All six were corrected, and the affected test cases were run again and passed.

Table 6.4: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	A new vendor registration was saved without a generated Vendor ID.	Vendor ID generation logic was missing.	Vendor ID generation added to the registration workflow (TC-01).	Closed

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-02	An uploaded document was saved without the Pending verification status.	The initial verification status was not set.	DS-02 now sets the verification status to Pending (TC-05).	Closed
BUG-03	The vendor status did not change after an approval decision.	The approval workflow did not update the related vendor.	DS-04 now updates the related vendor record (TC-11, TC-12).	Closed
BUG-04	The vendor did not receive the rejection e-mail.	The rejection notification was incorrectly configured.	The notification was reconfigured to use the vendor e-mail field (TC-16).	Closed
BUG-05	Documents about to expire were not included in the reminders.	The expiry reminder window was not checked correctly.	The date condition was corrected to cover today up to today plus 30 days (DS-05).	Closed
BUG-06	A vendor could access another vendor's records.	Record visibility restrictions needed correction.	Record visibility was corrected so that a vendor sees only its own records.	Closed

6.5 User Acceptance Testing

User acceptance testing was conducted by the project developer to evaluate the application from the perspective of the four user roles: Administrator, Vendor, Compliance Officer, and Procurement Manager. Each role was tested by performing its relevant tasks, including managing vendor records, submitting registration details, reviewing compliance documents, and recording approval decisions.

The testing confirmed that the dashboard provides a clear overview of vendor applications and compliance status, registration and document submission are straightforward, document review helps track pending and verified documents, and the approval process organizes vendor decisions and records. No changes were required based on this self-assessment

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator separates an application into Development, Stage and Production environments: changes are made and tried in Development and then published forward, so that real users work on a stable copy. The application was developed in the Development environment on the trial edition (the application header in Figure 5.1 shows "Trial expires in 14 days"). Publication to the Stage and Production environments is not evidenced in the supplied screenshots and is therefore recorded as not evidenced in Appendix A.

The deployment strategy for the application is as follows. Steps 1 to 3 are supported by the evidence; steps 4 to 6 are listed for completeness.

1. Develop and unit-test the forms, reports, dashboard and workflows in the Development environment.
2. Configure the roles, the permission profiles and the vendor record visibility (Section 7.3).
3. Test the notification workflows and the daily scheduled workflow with sample data (Chapter 6).
4. Publish the tested version to Stage and then to Production.
5. Share the application address with vendors and internal users and create their accounts.
6. Record release notes for each later version and repeat the publish from Development.

Table 7.1: Deployment Checklist

Step	Status	Evidence
Forms, reports, dashboard and workflows built in Development	Done	Chapter 5
Test cases run	Done	Table 6.1
Roles, permission profiles and record visibility configured	Partial	Figure 5.9; live role assignment not shown
Zoho Mail notification workflows configured	Partial	Figure 5.8; planned integration
Version published to Stage and Production	-	–
Tested with vendor and internal accounts in Production	-	–
Trial edition upgraded or application moved to a paid plan	Pending	About 14 days remained when the dashboard was captured

7.2 Performance Optimisation

No performance measurement was recorded, so no before-and-after figures are reported. The review of the scripts identified the opportunities in Table 7.2. They are proposals and have not been applied. The scheduled workflow loops over every compliance document each day; the remaining opportunities concern what the scripts fetch and when they run.

Table 7.2: Performance Optimisation Opportunities (Proposed, Not Measured)

Proposed Optimisation	Component	Current Approach	Proposed Approach	Expected Benefit
Fetch only documents in the window	Notify on Expiring Documents (DS-05)	Loops over every document and tests the dates inside the loop	Fetch only the documents whose expiry date lies between today and today plus 30 days	Fewer records processed as the document count grows
Avoid repeated vendor fetches	DS-05 and DS-03	One vendor fetch per document	Fetch each vendor once per run	Fewer record reads
Filter reports by default	Document Verification and All Vendors reports	List reports show all records	Add a default filter such as pending items or the current month	Smaller pages as data grows
Run the status update only on a decision	Update Vendor Approval Status (DS-04)	Runs on every created approval record	Keep the trigger on created records only	Avoids needless updates

7.3 Security Features

Table 7.3 lists the security controls and the evidence for each.

Table 7.3: Security Controls

Control	Implementation	Evidence / Gap
Authentication	Users sign in with their Zoho Creator accounts	The application header shows the signed-in user
Role design	Administrator, Vendor, Compliance Officer, Procurement Manager	Design in Table 4.7; live assignment not shown
Permission profiles	Approval Authority, Compliance Auditor, Risk Assessment Manager and Read	Figure 5.9; module-level settings not shown
Vendor record isolation	A vendor sees only its own records; internal reports and approval functions are not accessible to vendors	Corrected under BUG-06; record-level criteria not shown
Input validation	Mandatory fields, mobile number and PAN format validated on submission	TC-02 to TC-04
Document storage	Documents are stored in File Upload fields with the vendor record	Figure 5.2
Encryption and security testing	Not applied or not evidenced	Listed in Section 9.2

Personal and business data. The Vendor Registration form collects the company name, contact person, e-mail address, mobile number, address, PAN and GSTIN. The Field

Properties panel of Zoho Creator offers Data Privacy and Data Security options; they were not shown as selected in the screenshots, so field encryption is not claimed. **Notification sender.** The notification e-mails are sent from the administrator's address through the Zoho Mail workflows.

7.4 Governance and Administration

Governance in Zoho Creator is exercised from the console and from Settings, which offer Users, Permissions, Roles and Data Sharing. The supplied evidence does not show the audit trail, governance reports or metrics, so none is claimed, and the governance and audit trail feature is recorded as not implemented in Appendix A.

The following practices are recommended for the production deployment:

- **Review.** Review the roles, the permission profiles and the vendor record visibility at a fixed interval, and remove any redundant rule.
- **Test users.** Create one user per role so that restrictions can be tested after each change.
- **Test data.** Remove sample vendors, documents, checks and approval records before go-live and keep a separate test vendor for later trials.
- **Release notes.** Record each published version with the workflows and scripts that changed.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

A single scenario illustrates the working application. A vendor registers on the Vendor Registration form by entering its company and contact details, PAN and GSTIN. The application generates the Vendor ID, sets the status to Submitted and notifies the Compliance Officer by e-mail (Figure 5.8 shows the format of this notification). The vendor uploads its compliance documents with their issue and expiry dates; the application sets each to Pending and notifies the Compliance Officer. Figure 5.3 shows five stored documents, for example ISO_9001_Cert_2026.pdf with an issue date of 01-Nov-2026 and an expiry date of 01-Nov-2027, and Figure 5.1 shows that one document is Verified, three are Rejected and one is Pending.

The Compliance Officer verifies or rejects each document with remarks, and the vendor is notified. The Compliance Officer records compliance checks with a risk level; Figure 5.4 shows five checks of 02-Nov-2026, for example a PAN Verification with the result Pass and the risk level Low. The Procurement Manager then records the decision; Figure 5.5 shows five approval records and Figure 5.6 shows them grouped as two Approved, one Needs Correction and two Rejected. The vendor status is updated from the decision, and a daily reminder is sent for documents that expire within 30 days. The dashboard (Figure 5.1) shows the counts of vendors, documents, checks and approval records on one screen.

8.2 Achievement of Objectives

Table 8.1 maps the objectives of the project to the outcomes.

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Status
Design linked Vendor Registration, Compliance Documents, Compliance Checks and Approval Records forms	Four forms were built and the three other forms refer to the vendor through a lookup (Chapters 4 and 5)	Achieved
Automate Vendor ID, initial vendor status and initial document status	Vendor ID generation, DS-01 and DS-02 are in place; defects BUG-01 and BUG-02 were corrected (Chapter 6)	Achieved
Record verification, checks, risk levels, reviewers and remarks	The documents, checks and approval records carry their reviewer, remarks and dates (Figure 5.3, Figure 5.4, Figure 5.5)	Achieved
Record approval decisions and update the vendor status	The Approval Records form and DS-04 (TC-10 to TC-12; BUG-03 corrected)	Achieved

Objective	Outcome	Status
Notify by e-mail and send daily expiry reminders	DS-01 to DS-03 and DS-05 are configured; Zoho Mail workflows are the planned integration (Figure 5.8)	Achieved; integration partial
Provide reports and a dashboard	Reports and the dashboard are in place (Figure 5.1, Figure 5.3 to Figure 5.6); three reports and three KPIs are not visible in the screenshots	Achieved;
Control access by role	Four roles and permission profiles are designed (Figure 5.9); live role assignment is not shown (BUG-06 corrected)	achieved

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing (Manual) System	Vendor Onboarding and Compliance Management System
Vendor records	Spreadsheets, e-mails and folders	Four linked forms in one application
Vendor identification	By name, copied between files	Generated unique Vendor ID
Documents	Files in separate folders without dates	File Upload field with issue date, expiry date and verification status
Review	Notes in e-mails	Compliance Checks form with result, risk level, reviewer and remarks
Decision	E-mail from a manager	Approval Records form that updates the vendor status
Notification	Manual e-mails	Workflow e-mails to the Compliance Officer and to the vendor
Expiry tracking	Personal calendar reminders	Daily scheduled reminder for documents expiring within 30 days
Summary view	Manual count	Reports and a dashboard
Access control	Physical possession of the files	Roles, permission profiles and vendor record visibility

8.4 Limitations of the Low-Code Approach

Several limitations were met in this project. First, platform behaviour that is easy to configure can be hard to inspect. The defects BUG-03 and BUG-04 show that event-driven logic depends on knowing exactly which record and which field a script reads, which matches the finding that event handling is a difficult area of low-code development [5]. Second, record-level access had to be corrected after testing (BUG-06), which shows that visibility settings must be tested with real accounts. Third, the Vendor lookup of Compliance Documents is configured with the display type Multi Select, so the platform allows a record to name several vendors unless the setting is restricted. Fourth, the expiry reminder is a simple date window and sends a reminder every day within it. Fifth, the trial edition limits the time in which the application can be demonstrated. Finally, as practitioners also report,

customisation beyond the built-in components is limited [4]; features such as a separate REST API integration would need a pro-code extension, and they are outside the scope of this project.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code configuration: the forms, the lookups, the reports, the Kanban board, the calendar and the dashboard. The status updates, the notifications and the expiry reminder needed short Deluge scripts, which is the low-code part. No pro-code component was needed (Section 5.10). The project shows that a citizen developer with basic scripting skills can build a working compliance workflow application, and that a professional developer would be needed for features that go beyond the platform.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Vendor Onboarding and Compliance Management System demonstrates a working low-code solution for vendor onboarding and compliance management on Zoho Creator. The application (Vendor Compliance System) provides four forms (Vendor Registration, Compliance Documents, Compliance Checks and Approval Records), six reports, a dashboard with KPI tiles and charts, Deluge workflows for status updates and notifications, a daily scheduled workflow for expiring documents, a vendor status process and a role design for four kinds of user. A vendor receives a unique Vendor ID, documents are verified, compliance checks and approval decisions are recorded against the vendor, and the vendor status follows the decision. All sixteen test cases passed and six defects were corrected.

The work also has limits that are recorded openly in Appendix A: the live role configuration and vendor-specific access, the Zoho Mail integration, publication across environments, security testing and the audit trail are only partly evidenced or not evidenced. These items should be closed and re-tested before the application is relied upon in a live procurement process.

9.2 Future Enhancements

- Capture and confirm the live assignment of users to the four roles and a sign-in test for each role.
- Restrict the Vendor lookup of Compliance Documents to a single vendor per record.
- Publish the tested version through the Stage and Production environments and record the release.
- Configure the vendor portal and the mobile layout for vendors.
- Configure the approval process and the vendor status flow as separate Creator components.
- Mark vendor identity and tax fields as personal data and consider field encryption.
- Enable the audit trail and review the governance reports of the application.
- Measure the performance of the reports and the scheduled workflow and apply the optimisations of Table 7.2.
- Add charts such as vendors by risk level to the dashboard.

9.3 Learning Outcomes and Career Relevance

The project exercises the main skills of the course: modelling data with forms and lookups, validations, building list, calendar, Kanban and chart reports and a dashboard, scripting with Deluge, configuring form workflows and a scheduled workflow, setting up notifications, designing roles, permission profiles and record visibility, and reasoning about deployment across environments. It also shows the importance of checking automation against real records, as the defects in Table 6.4 demonstrate. These skills are relevant to roles in business-application development, ERP and workflow automation, and low-code or citizen-development support.

REFERENCES

[1] Zoho Corporation, "Zoho Creator Resource Center." [Online]. Available: https://www.zoho.com/creator/help/

[2] Zoho Corporation, "Introduction to Deluge," Zoho Deluge Help. [Online]. Available: https://www.zoho.com/deluge/help/

[3] Zoho Corporation, "Zoho Mail Help Center." [Online]. Available: https://www.zoho.com/mail/help/

[4] Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, "Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective," in Proc. 15th ACM/IEEE Int. Symp. Empirical Software Engineering and Measurement (ESEM), 2021. [Online]. Available: https://arxiv.org/abs/2107.07482

[5] M. A. Al Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider and A. Iqbal, "An Empirical Study of Developer Discussions on Low-Code Software Development Challenges," arXiv:2103.11429, 2021. [Online]. Available: https://arxiv.org/abs/2103.11429

[6] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, "Supporting the understanding and comparison of low-code development platforms," in *Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA)*, 2020, pp. 171–178.

[7] R. Waszkowski, "Low-code platform for automating business processes in manufacturing," *IFAC-PapersOnLine*, vol. 52, no. 10, pp. 376–381, 2019.

[8] Zoho Corporation, "Deluge Script Reference Guide." [Online]. Available: https://www.zoho.com/deluge/ (Accessed: Oct. 19, 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented, Partial = implemented in part, N = not implemented. The checklist reflects the supplied evidence; examiners should verify the entries against the live application.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Fig. / Code
I	Application created and menu organised	Y	Dashboard; Vendor Registration; Compliance Documents; Compliance Checks; Approval Records	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Four main forms	5.2	Fig. 5.2; Table 4.2 to Table 4.5
II	Lookup fields / relationships	Y	Vendor lookups in Documents, Checks, and Approval Records	4.2, 5.2	Fig. 4.2; Fig. 5.2
II	Subform	N	Not used	4.2	—
II	Field validations / layout rules	Y	Mandatory fields and input validations; layout rules not evidenced	5.2	Fig. 5.2
II	List report with filter, grouping, custom action	Partial	Vendor and compliance reports; advanced grouping/custom actions not evidenced	5.3	Fig. 5.3; Fig. 5.4; Fig. 5.5
II	Kanban / Calendar / Timeline / Map report	Partial	Approval status board and expiry calendar planned	5.3	Fig. 5.6
II	Pivot table / chart report	Y	Dashboard charts	5.4	Fig. 5.1
II	UI Builder dashboard	Y	Vendor Compliance Dashboard	5.4	Fig. 5.1
II	Deluge form events	Y	Registration, document verification, approval workflows	5.5	Code C.1–C.4
II	Deluge functions, fetch, loops, collections	Y	Expiry workflow loop; custom functions not evidenced	5.5	Code C.5
II	Email / in-app notifications	Y	Registration, upload, verification, expiry notifications	5.5, 5.6	Code C.1–C.5
II	Form workflow	Y	Vendor Registration, Compliance Documents, Approval Records	5.6	Fig. 5.7
II	Scheduled workflow	Y	Daily document expiry reminders	5.6	Code C.5
II	Approval process	Y	Approval Records form and decision-based status update	5.6	Fig. 5.7; Code C.4
II	Blueprint	Partial	Vendor status flow planned	5.6	Fig. 4.5
II	Batch / report workflow	N	Not used	5.6	—
II	Testing and debugging evidence	Y	Test cases and debugging observations	6	Table 6.1–Table 6.3
III	Integration with a Zoho service	Y	Zoho Mail notifications planned	5.7	Code C.1–C.5

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Fig. / Code
III	Connection / REST API (invokeurl)	N	No REST API call evidenced	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles, permission profiles	Y	Four roles planned; live configuration not evidenced	5.8	Fig. 5.9
III	Record-level access / portal	Y	Vendor-specific access	5.8	Table 5.8
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics	Y	Dashboard KPI cards and charts; Zoho Analytics not used	5.9	Fig. 5.1
IV	Performance optimisation applied and measured	N	No measurements documented	7.2	—
IV	Deployment across environments	N	Not evidenced	7.1	—
IV	Security controls	Y	Role and permission design; encryption/security testing not evidenced	7.3	—
IV	Governance / audit trail	N	Audit trail not evidenced	7.4	—
V	Pro-code extension	N	Not used	5.10	—
V	Low-code limitations and spectrum analysis	Y	Discussed in report	8.4, 8.5	—
V	Mobile app / portal access	Y	Vendor portal planned;	5.11	—

Coverage summary: 20 fully implemented,4 partially implemented and 8 not implemented, out of 32 features.

APPENDIX B – COURSE OUTCOME MAPPING

The self-assessed level (1 – Low, 2 – Medium, 3 – High) is left for the student; the examiner may record a level in the last column.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	5	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	5	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9, 7.3	5	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	4	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	5	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The scripts below are reproduced with their comments and purposes. Field link names and form link names are those of the application configuration.

C.1 Workflow "Notify on New Vendor Registration" (Vendor Registration form)

Code C.1: DS-01 Notify on New Vendor Registration

```
// Script ID : DS-01
// Location  : Vendor Registration -> Created -> Successful form submission
// Purpose   : Set initial vendor status and notify the Compliance Officer

input.vendor_status = "Submitted";

sendmail
[
    from : zoho.adminuserid
    to : "compliance@yourcompany.com"
    subject : "New Vendor Registration - " + input.company_name
    message : "A new vendor registration has been submitted.\n\n"
            + "Vendor Name: " + input.company_name + "\n"
            + "Contact Email: " + input.email_address + "\n"
            + "Please review the application in the Vendor Compliance System."
];
```

C.2 Workflow "Notify on Document Upload" (Compliance Documents form)

Code C.2: DS-02 Notify on Document Upload

```
// Script ID : DS-02
// Location  : Compliance Documents -> Created -> Successful form submission
// Purpose   : Set document status to Pending and notify the Compliance Officer

input.verification_status = "Pending";

vendorRecord = vendor_registration[ID == input.vendor];

sendmail
[
    from : zoho.adminuserid
    to : "compliance@yourcompany.com"
    subject : "Document Uploaded for Verification"
    message : "A new compliance document has been uploaded.\n\n"
            + "Vendor: " + vendorRecord.company_name + "\n"
            + "Document Type: " + input.document_type + "\n"
            + "Please review the uploaded document."
];
```

C.3 Workflow “Notify Vendor on Document Verification” (Compliance Documents form)

Code C.3: DS-03 Notify Vendor on Document Verification

```
// Script ID : DS-03
// Location  : Compliance Documents -> Edited -> On update of Verification Status
// Purpose   : Notify vendor when a document is verified or rejected

vendorRecord = vendor_registration[ID == input.vendor];

if(vendorRecord.email_address != null && vendorRecord.email_address != "")
{
    sendmail
    [
        from : zoho.adminuserid
        to : vendorRecord.email_address
        subject : "Document Verification Update"
        message : "Dear " + vendorRecord.company_name + ",\n\n"
                + "Your " + input.document_type + " has been reviewed.\n"
                + "Verification Status: " + input.verification_status + "\n"
                + "Reviewer Remarks: " + input.reviewer_comments + "\n\n"
                + "Please log in to the Vendor Compliance System for details."
    ];
}
```

C.4 Workflow “Update Vendor Approval Status” (Approval Records form)

Code C.4: DS-04 Update Vendor Approval Status

```
// Script ID : DS-04
// Location  : Approval Records -> Created -> Successful form submission
// Purpose   : Update vendor status based on the approval decision

vendorRecord = vendor_registration[ID == input.vendor];

if(input.decision == "Approved")
{
    vendorRecord.vendor_status = "Approved";
}
else if(input.decision == "Rejected")
{
    vendorRecord.vendor_status = "Rejected";
}
else if(input.decision == "Needs Correction")
{
    vendorRecord.vendor_status = "Needs Correction";
}

vendorRecord.verified_by = zoho.loginuser;
vendorRecord.submission_date = zoho.currentdate;
```

C.5 Scheduled Workflow “Notify on Expiring Documents” (Compliance Documents form)

Code C.5: DS-05 Notify on Expiring Documents

```
// Script ID : DS-05
// Location  : Compliance Documents -> Scheduled workflow -> Daily
// Purpose   : Notify vendors about documents expiring within 30 days

for each doc in vendor_document
{
    if(doc.expiry_date != null
       && doc.expiry_date >= zoho.currentdate
       && doc.expiry_date <= zoho.currentdate.addDay(30))
    {
        vendorRecord = vendor_registration[ID == doc.vendor];

        if(vendorRecord.email_address != null
           && vendorRecord.email_address != "")
        {
            sendmail
            [
                from : zoho.adminuserid
                to : vendorRecord.email_address
                subject : "Compliance Document Expiry Reminder"
                message : "Dear " + vendorRecord.company_name + ",\n\n"
                        + "Your " + doc.document_type
                        + " is due to expire on " + doc.expiry_date + ".\n"
                        + "Please upload an updated document before the expiry
date."
            ];
        }
    }
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the Vendor Compliance System at https://creatorapp.zoho.in/nithiyasreenc/vendor-compliance-system and sign in with the account assigned to your role. The menu on the left shows Dashboard, Compliance Documents, Approval Records, Compliance Checks and Vendor Registrations. The application can be opened in a desktop, tablet or phone browser; the Zoho Creator mobile app has not been configured.

D.2 Instructions for the Vendor

1. Open Vendor Registrations and complete the Vendor Registration form with the company name, contact person, e-mail address, mobile number, address, state, PIN code, vendor type, PAN and GSTIN, then select Submit.
2. Note the Vendor ID generated for your application; your status is Submitted.
3. Open Compliance Documents and add each required document: choose your company, the document type, attach the file and enter the issue date and the expiry date, then select Submit.
4. Read the e-mails “Document Verification Update” for the status of each document and the reviewer remarks.
5. If a document is rejected or the status is Needs Correction, update the details or upload a corrected document.
6. Upload an updated document before the expiry date when you receive the reminder “Compliance Document Expiry Reminder”.

D.3 Instructions for the Compliance Officer

1. Open the Pending Applications report to see the applications waiting for review, and set the vendor status to Under Review when you start.
2. Open the Document Verification report, open each document, change the Verification Status from Pending to Verified or Rejected and enter the Reviewer Remarks.
3. Open Compliance Checks, add a check with the vendor, check type, result, risk level, reviewer, check date and remarks, and submit.
4. Request corrections from the vendor when documents or checks are incomplete.
5. Open the Expiring Documents calendar to see the documents that will expire soon.

D.4 Instructions for the Procurement Manager

1. Open the Approval Pipeline board and the compliance results of the vendor to be decided.
2. Open Approval Records, select the vendor, enter the reviewer, choose the decision (Approved, Rejected or Needs Correction), enter the comments and the decision date, and submit.
3. Check that the vendor status has changed and that the vendor has been notified.

D.5 Instructions for the Administrator

1. Open Settings → User Permissions to review the roles, the permission profiles and the record visibility.
2. Open the Dashboard to see the KPI tiles, the Verification Status block and the charts, and use the Quick Links to add a vendor registration, a compliance document, a compliance check or an approval record.
3. Open the reports All Vendors, Pending Applications and Document Verification to review the records.
4. Review the workflows under Workflow → Form workflows and the scheduled workflow for expiring documents.